

The State of AI Financing: A Pause, Not a Pullback

September marked the second-lowest month of AI-related supply this year, with only ~\$23bn globally. In our view, this slowdown is temporary and driven by a combination of significant front-loading of issuance in the year and building political/regulatory constraints. We expect a busier fall and 4Q, as the focus shifts to 2027 capex.

Key Takeaways

- AI issuance underwhelmed in September, in line with our expectations. However, this does not reflect weakening fundamentals or shortage of capital to fund the buildout. In our view, September's supply slowdown is due to a combination of 1) significant front-loading of issuance that happened in previous months, especially over the summer, 2) emerging signs of regulatory/political challenges to new data center construction, and 3) the rates sell-off possibly requiring project-level renegotiations for data-center deals.
- In US IG, we think the macro backdrop - and not elevated supply - is the biggest test between now and year-end, with outflows expected on the back of a sharp drawdown in returns. We expect at least one more deal in the US market, and note that HQ hyperscalers continue to tap non-USD currencies in between their larger US deals earlier this year. HQ hyperscalers should remain largely insensitive to higher rates, and we expect another wave of volatility into the fall, especially around 3Q26 earnings and likely further upward capex revisions.
- Lev Fin was active in September, but we expect 4Q26 to look more like 1Q/3Q, with a slower pace of deals given higher rates and regulatory/political limitations. Lower-quality issuers and project-level debt are more sensitive to the current rate environment, in our view, but issuers might be able to negotiate better lease economics with tenants to improve their returns.
- A handful of new securitized products deals came to market in September after a pause in August, bringing with them wider spreads and structural innovations. A floating-rate ABS deal and partial amortization in CMBS reflect the evolution the market is undergoing to attract incremental demand as the cost of debt for sponsors surges. We expect spreads to continue to leak wider from here, and reiterate our preference for colocation over hyperscaler data center ABS. Raising interest rate volatility could adversely affect the economics of data center securitization.
- For Single Name, despite lighter September issuance, off-cycle earnings and headline flow continue to drive elevated investor inbound. Both ORCL and NVDA, the latter a new addition to our coverage, remain central to these discussions, for their evolving funding strategies and their contribution to heightened lease documentation scrutiny.

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Tech Diffusion

A Morgan Stanley Research
Key Theme of 2026

With ~\$466bn of issuance YTD, we think 4Q supply might slow down sequentially, while remaining above September's issuance pace.

Different from last year, when most of the supply came in 4Q, we expect a more moderate pace of issuance in 4Q26, with different dynamics in each market: we expect hyperscalers to come back to the US IG market and continue to tap non-USD currencies; supply of data center paper in both the IG

and HY markets is hard to predict as there is certainly urgency to get construction going, but there is also more sensitivity to higher rates. We also lowered our 2026 issuance estimates for ABS/CMBS to \$25-30bn, which implies another ~\$5-10bn of supply in the upcoming quarter.

Collateral-backed paper still has better risk-reward than unsecured corporates, in our view. Within the universe of collateralized assets (~30% of AI-related debt), construction risk on data center paper in the IG/HY markets has been a significant driver of spread volatility on the back of headlines about permit and power delays, as well as the risk of lease reassignment and contract uncertainties ([ORCL's Force Majeure episode](#)). Stabilized assets in ABS/CMBS are therefore more insulated from such negative volatility. As for chip financing, we think the nature of the assets allows for faster cash flow generation and amortization, but there is still only a handful of public deals that would allow investors to express this preference.

Despite better performance last month from the hyperscalers, we expect near-term volatility amid rising rates/outflows going into the midterm elections and likely another leg of upward capex revision at 3Q26 earnings. Looking past 4Q26, however, we think much of the hyperscaler decompression trade may have played out: issuance cadence is becoming more (although not completely) predictable, capex is shifting to shorter-dated assets (like chips), and continued evidence of strong returns on AI capex underscore that the fundamental story behind the spending remains robust.

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Other relevant content on AI financing:

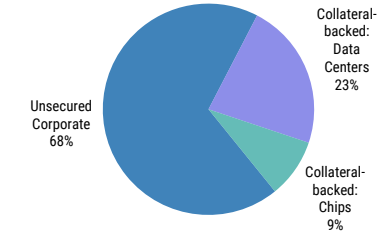
- [Securitized Products & US Credit Strategy: Data Center Financing: Funding Stabilized Assets in a Higher Rate Environment \(24 Sep 2026\)](#)
- [TMT Credit, Clean Tech Energy, Corporate/Securitized Credit Strategy: Project Jupiter Drifts Off Orbit – The Gravity of Force Majeure \(25 Sep 2026\)](#)
- [TMT Credit Research: NVDA Credit: Balance Sheet-as-a-Service \(24 Aug 2026\)](#)
- [Sunday Start | What's Next in Global Macro: The Politics of Compute \(13 Sep 2026\)](#)

Exhibit 1: 2026 YTD issuance is now at \$466bn, vs. \$216bn in FY25 issuance

Issuance (USD bn)	YTD-26	YTD-25	FY25A
HQ Hyperscalers	132	5	68
Oracle	25	26	26
SpaceX	25	-	-
Hyperscaler Total	182	31	93
Secd DC Const. (144A)	43	-	27
Semis + REITs	46	34	35
4a2 Priv. Placement	35	-	-
Total IG	306	65	156
Total HY	53	7	15
Secd Lev. Loans	10	2	2
Total US Corporate Credit	369	73	173
Total US Securitized Credit	20	19	27
Non-USD Corp. Credit	75	8	15
Non-USD Securitized Credit	2	1	1
Total Non-USD	77	8	16
Total Global Credit Markets	466	101	216

Source: Bloomberg, Pitchbook | LCD, CreditFlow, TREPP, Morgan Stanley Research estimates. Note: YTD 2026 data includes deals priced until 9/30/26. For a more detailed version of this exhibit, refer to [Exhibit 3](#) . 4a2 Priv. Plac. includes only the AVGO chip financing deal. Figures are rounded to the nearest \$bn; totals are calculated using unrounded figures and may differ slightly from the sum of displayed values.

Exhibit 2: ~70% of AI-related debt is unsecured corporate issuance



Source: Bloomberg, Company filings, Morgan Stanley Research. For data center breakdown and rel val, see [Exhibit 9](#) and [Exhibit 5](#) .

Issuance Summary & September Pricing

Exhibit 3: We're now at \$466bn of AI-related debt issuance across global credit markets

Markets (USD bn figures)		Issuers	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	2026 YTD	2025 YTD	Δ%
Corporate Credit (U.S.)	U.S. Investment Grade	HQ Hyperscalers	-	20	37	25	-	-	25	25	-	132	5	2540%
		Oracle	-	25	-	-	-	-	-	-	-	25	26	-3%
		SpaceX	-	-	-	-	-	25	-	-	-	25	-	-
		Semiconductors	5	-	-	8	-	25	-	5	-	42	34	23%
		REITs	-	2	-	-	-	-	3	-	-	5	-	-
		Secd DC Const. (144A)	-	-	-	22	-	4	13	4	-	43	-	-
		4a2 Private Placement	-	-	-	-	-	35	-	-	-	35	-	-
		Total IG	5	47	37	54	-	89	41	34	-	306	65	374%
	U.S. Leveraged Finance	Secd DC Const.	-	6	2	16	-	4	4	2	3	37	-	-
		Corporate	-	-	-	3	-	3	-	-	10	16	7	130%
		Secd Loans	-	-	-	-	3	-	3	4	1	10	2	419%
Total LevFin		-	6	2	19	3	7	7	6	14	63	9	619%	
Total U.S. Corporate Credit		5	52	39	73	3	96	47	39	14	369	73	403%	
Securitized Credit (U.S.)	Securitized Credit	ABS	1.9	2.9	2.7	1.4	-	2.3	0.9	-	1.5	14	12	19%
		CMBS	-	2.1	-	-	1.1	1.8	1.3	-	0.4	7	8	-15%
	Total U.S. Securitized Credit		2	5	3	1	1	4	2	-	2	20	19	5%
Total U.S. Credit Markets			6	57	42	74	4	100	50	39	16	389	93	320%
Corporate Credit (non-USD) ¹	IG Hyperscalers	EUR	-	-	17	-	11	-	-	-	-	27	8	255%
		GBP	-	8	-	-	-	-	-	-	6	13	-	-
		CAD	-	-	-	-	6	10	-	-	-	16	-	-
		CHF	-	4	-	-	4	-	-	-	-	8	-	-
		JPY	-	-	-	-	4	-	-	-	-	4	-	-
		AUD	-	-	-	-	-	-	-	4	-	4	-	-
	Total IG Other Currencies		-	11	17	-	24	10	-	4	6	72	8	836%
Total HY Other Currencies		-	-	-	-	-	2	-	-	1	3	-	-	
Total Non-USD Corporate Credit		-	11	17	-	24	12	-	4	7	75	8	881%	
Securitized Credit (non-USD) ¹	Securitized Credit	ABS	-	0.7	-	-	-	-	-	-	-	1	1	-1%
		CMBS	-	-	0.8	-	-	-	0.4	-	-	1	-	-
	Total Non-USD Securitized Credit		-	1	1	-	-	-	0	-	-	2	1	165%
Total Non-USD Credit Markets			-	12	18	-	24	12	0	4	7	77	8	820%
Total Global Credit Markets			6	69	59	74	28	113	50	43	23	466	101	362%

Notes: Includes transactions priced until 9/30/26. Presented figures exclude IG TLs such as AMZN DDTL. 1. All figures in USD-equivalent billions.

Source: Bloomberg, PitchBook | LCD, CreditFlow, TREPP, Morgan Stanley Research estimates. Notes: YTD 2026 data includes deals priced until 9/30/26. 4a2 Private Placement includes only the AVGO chip financing transaction. Figures are rounded to the nearest \$bn; totals are calculated using unrounded figures and may differ slightly from the sum of displayed values. Δ% is YTD vs. YTD and uses unrounded figures.

Exhibit 4: Deals priced in September have generally tightened since issuance

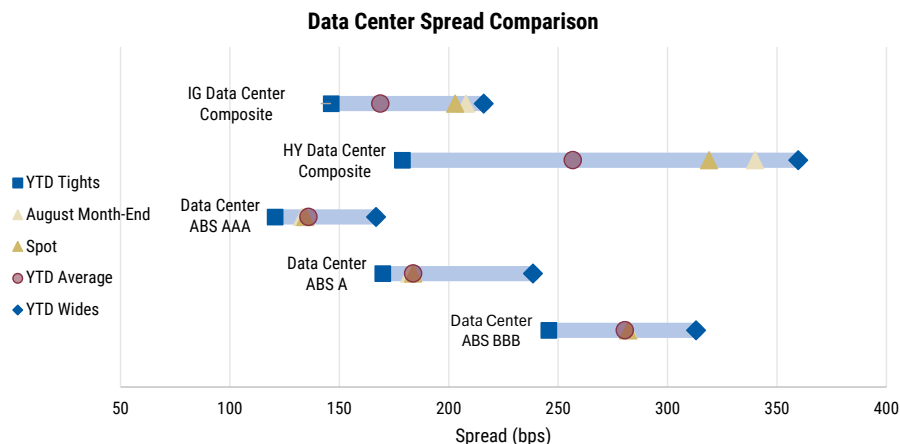
Ticker	Issue Date	Maturity Date	Size (\$bn)	Composite Rating	Coupon	Issue Spread to Benchmark (bps)	Current Spread to Benchmark (bps)	Spread Δ Since Issuance (bps)
Corporate Credit - High Yield								
CLSPRK	9/18/2026	10/1/2031	2.28	BB-	7.88%	338	303	-35
SOFTBK	9/23/2026	4/1/2030	1.00	BB+	8.63%	363	292	-71
SOFTBK	9/23/2026	4/1/2032	4.50	BB+	9.25%	422	398	-24
SOFTBK	9/23/2026	4/1/2034	4.50	BB+	9.75%	467	428	-39
DDRIVE	9/23/2026	9/30/2031	1.12	BB-	8.88%	422	482	+60
Corporate Credit - Leveraged Loans								
Crusoe (S+425)	9/23/2026	11/29/2030	0.51	N.R. ¹	8.02%	433	409	-24
Corporate Credit - Investment Grade Reverse Yankee								
AMZN (GBP)	9/9/2026	9/14/2032	1.35	AA-	5.55%	75	70	-5
AMZN (GBP)	9/9/2026	9/14/2038	1.35	AA-	6.25%	90	86	-4
AMZN (GBP)	9/9/2026	9/14/2045	1.35	AA-	6.65%	93	86	-7
Corporate Credit - High Yield Non-USD								
SOFTBK (EUR)	9/23/2026	10/1/2030	0.57	BB+	7.13%	373	388	+15
SOFTBK (EUR)	9/23/2026	10/1/2032	0.57	BB+	8.00%	456	468	+12
Securitized Credit - ABS²								
VDC 2026-1	9/11/2026	3/15/2032	0.23	AAA	6.13%	140	142	+2
VDC 2026-1	9/11/2026	3/15/2032	0.11	AA-	6.52%	180	177	-3
VDC 2026-1	9/11/2026	3/15/2032	0.21	A	6.72%	200	194	-6
QTSII 2026-8 A2-I	9/30/2026	10/5/2031	0.32	AAA	6.44%	150		
QTSII 2026-8 A2-II	9/30/2026	10/5/2031	0.06	AA-	6.93%	200		
QTSII 2026-8 A2-III	9/30/2026	10/5/2031	0.16	A	7.12%	250		
QTSII 2026-8 B	9/30/2026	10/5/2031	0.13	BBB-	8.38%	355		
QTSII 2026-9 A2-II	9/30/2026	10/5/2033	0.04	AA-	7.15%	215		
QTSII 2026-10 A2-III	9/30/2026	10/5/2031	0.22	AAA		SOFR+175		
QTSII 2026-11 A2-III	9/30/2026	10/5/2030	0.08	AAA				
Securitized Credit - CMBS²								
DATA 2026-800D A	9/8/2026	9/11/2031	0.32	AAA	6.42%	200	179	-21
DATA 2026-800D B	9/8/2026	9/11/2031	0.03	AA-	7.00%	260	237	-23

Notes: 1. Crusoe TL (borrower IS An03 LLC) was rated Ba1 by Moody's and (P) BB by S&P. 2. For ABS and CMBS, maturity date reflects the anticipated repayment date (ARD).

Source: Bloomberg, CreditFlow, Green Street, Morgan Stanley Research. Current levels as of 9/30/26. We reflect select tranches only for some of the deals in this table.

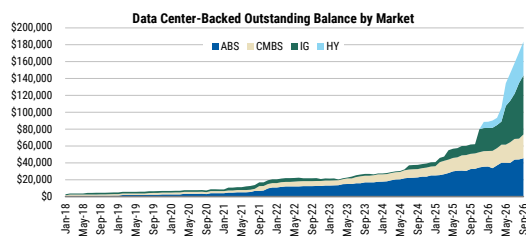
Data Center Relative Value: Corporate Credit vs. Securitized Credit

Exhibit 5: Data center-backed paper in corporate credit widened over the last six weeks, while ABS paper remained relatively steady



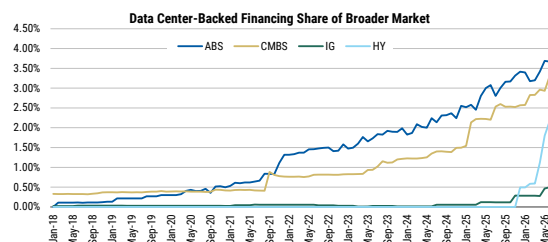
Source: Bloomberg, Morgan Stanley Research. Spot levels as of 9/28/26.

Exhibit 6: Issuance of DC-backed paper has ramped quickly across markets this year, with the biggest gains in IG and HY



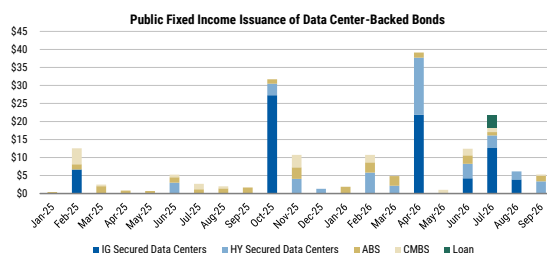
Source: Bloomberg, CreditFlow, Trepp, Intex, Morgan Stanley Research

Exhibit 7: Despite heavy supply all year, as a share of the broader markets DC-backed paper remains a very small proportion



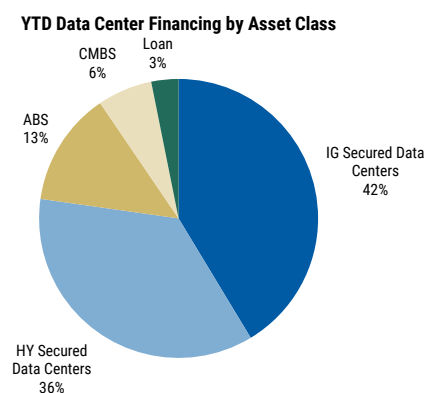
Source: Bloomberg, CreditFlow, Trepp, Intex, Morgan Stanley Research. Data through Sept 1, 2026.

Exhibit 8: Monthly issuance of data center-backed paper by product



Source: Bloomberg, CreditFlow, Trepp, Morgan Stanley Research

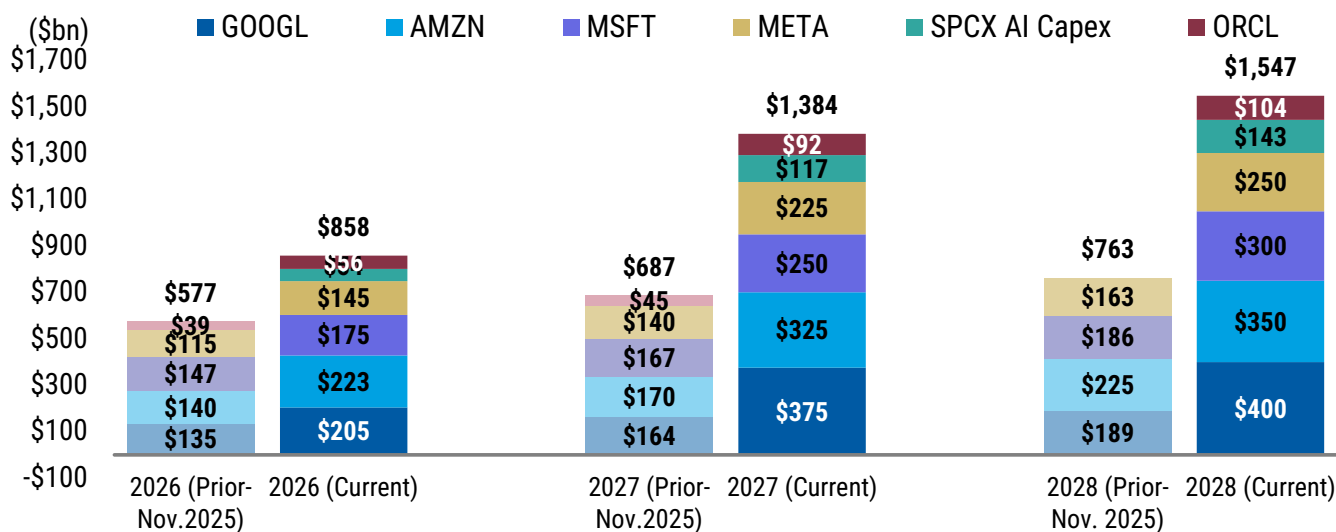
Exhibit 9: YTD financing breakdown of data center-backed paper



Source: Bloomberg, CreditFlow, Trepp, Morgan Stanley Research

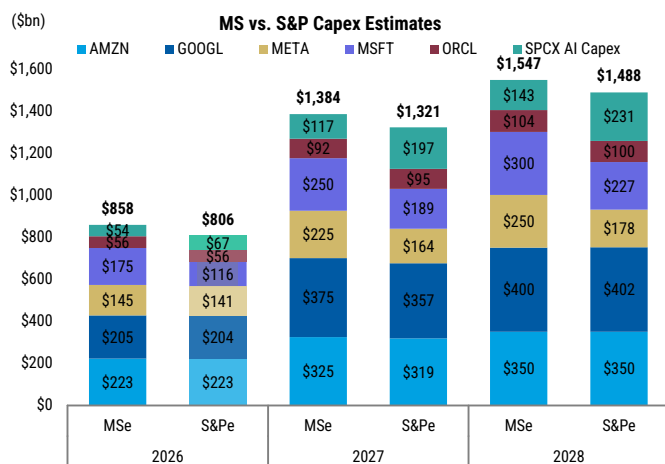
2027 Hyperscalers' Capex Estimates Are At ~ \$1.4tr

Exhibit 10: We expect another cycle of capex revisions next month as we go into 3Q26 earnings



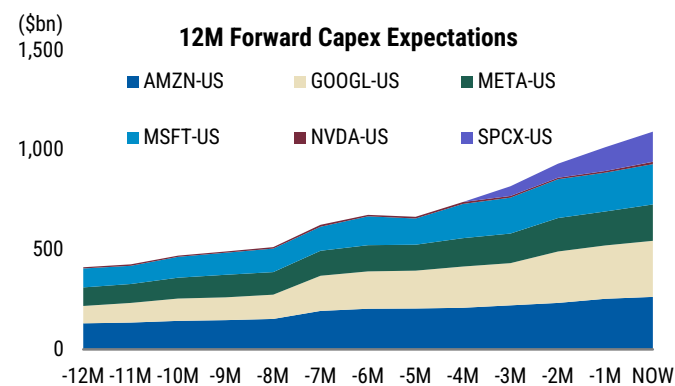
Source: Morgan Stanley Research current estimates from models dated as of: 7/31/26 for AMZN, 8/24/26 for GOOGL, and 7/30/26 for META (covered by Brian Nowak); 8/9/26 for MSFT, 9/11/26 for ORCL, and 9/15/26 for SPCX (covered by Adam Wood, Sanjit Singh, and Adam Jonas respectively). Prior estimates are from November 2025. Company-level capex figures are rounded to the nearest \$bn; total capex is calculated using unrounded figures and may differ slightly from the sum of displayed labels.

Exhibit 11: S&P capex estimates are in line with MS estimates



Source: S&P, Morgan Stanley Research estimates

Exhibit 12: NTM consensus capex expectations are +8% from last month



Source: FactSet, Morgan Stanley Research

US Investment Grade Credit: Macro Poses a Bigger Test Than Elevated Supply

There was no AI-related issuance in the US IG market during the month of September. Following a busy summer, we were expecting a slowdown in near-term supply, considering how much had been done already and issuers' efforts to better establish cadence. Importantly, we think it's now fairly well understood that higher capex next year will likely come with higher issuance. On the other hand, the macro backdrop has become more challenging, with rates selling off sharply and a more hawkish Fed path priced in.

Therefore, thinking through the upcoming weeks, the macro environment will likely be the bigger test for corporate credit markets, while issuance might take a backseat.

In our view, the credit market at large should be able to handle the sell-off in yields.

With nominal growth running at an impressive 6%+ and earnings growth even better, the rate move has a strong fundamental underpinning and as such, we expect credit quality to hold up really well. Importantly, AI-related capex has been a [large anchor for US growth](#), and our economists see it being relatively insensitive to the level of yields, at least locally. Notably, our economists' rate path is more dovish than consensus, calling for [two more rate hikes](#) (vs. market pricing that is closer to four).

Technical factors will likely dominate in the near term, especially around investor outflows.

With IG down 3% in total return on the year and 4% on the quarter, the drawdown has reached a level where mutual funds/ETFs could see meaningful redemptions. But it is worth remembering that over the past five years, the credit market has seen multiple bouts of rate repricing, as the 10Y has gone from 1% to over 5%. While the rate-hiking cycle of 2021/2022 created a sustained drag on spreads and outflows, that was a case of much higher inflation and the Fed being behind the curve. Today it appears to be a case of inflation not coming down fast enough, coupled with robust growth driving a modest reset higher in policy rates.

When it comes to debt-financing future capex, **we expect high-quality hyperscalers to remain largely insensitive to higher rates**, given strong visibility into attractive returns on invested capital (for those interested in returns math, we recommend slides 22-29 in [this deck](#)) and debt still proving to be a cheaper funding source relative to equity. While the equity market is not rewarding the 'capex factor' to the same extent as earlier in the year, our [equity analysts](#) expect strong capex growth next year, supported by continued demand for compute, incrementally expanding inference workloads, and enterprise adoption.

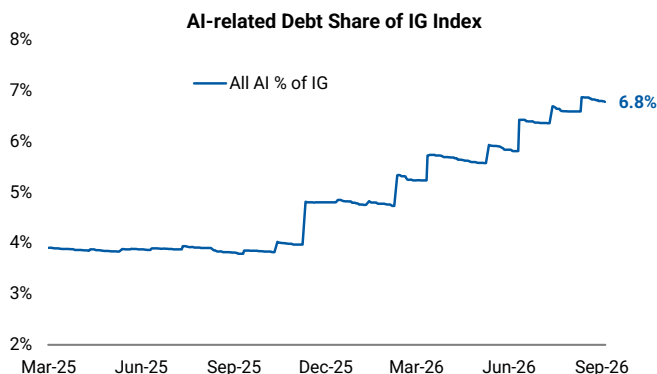
Near term we expect continued chopiness going into the midterm elections and 3Q26 earnings, with arguably another leg of upward capex revision (consensus is still at ~\$1-1.1tn for 2027 combined hyperscaler capex, MSe is at ~\$1.4tn). As for the lower-rated cohorts (see also [Lev Fin section](#)), we expect more sensitivity to higher cost of capital, and continue to see more spread volatility on headlines around construction delays and political pushback to build-outs. Finally, we think relative value will play a part for the BBB cohort given TMT-related supply outside of the AI players (~\$35bn of BBB TMT debt has been issued since August).

Looking ahead beyond 4Q26, however, we think that much of the hyperscaler decompression trade has played out. While a persistent basis to the broader index is warranted on elevated absolute and relative supply, spreads have widened a fair amount (similar to banks in the past). There is more (albeit not perfect) clarity on the cadence and frequency of issuance. Our analysts' estimates for ROIC also remain robust, with more evidence of that coming through over the next year.

We also think that in 2027, capex will shift toward components such as chips, where financing is more tailored to short duration/asset-level/private capital. That doesn't mean less IG issuance per se, but potentially a smaller y/y increase than we saw this year. Additionally, as rates stabilize, yield-based investors will likely continue to remain an anchor (despite low spreads) due to the moves on the Treasury benchmark. The story is arguably different for the big semiconductor names, in our view, as we've gotten little clarity so far on the magnitude and format of chip financing structures. A lot of the chip financing remains ahead of us, and what form it will take/which markets and investor pockets will be involved are all unknowns for now (see the [recent credit initiation on NVDA](#) for more thoughts).

AI-Related Issuance vs. IG Market Composition

Exhibit 13: AI-related debt share of the IG Index continues to rise rapidly on the back of heavy issuance



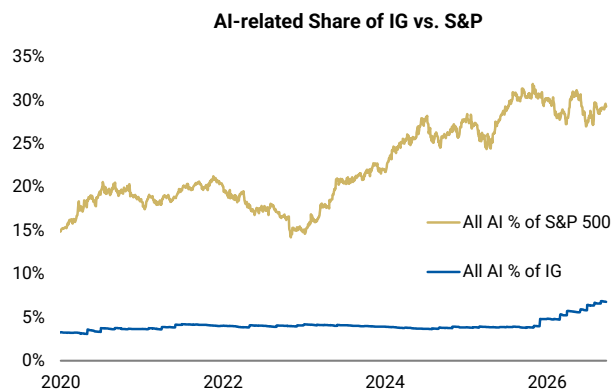
Source: Bloomberg, Morgan Stanley Research. Note: AI-related includes a broader universe than just hyperscalers.

Exhibit 15: Hyperscaler index debt adds up to \$448bn, "AI enabling" semi names add \$87bn

Issuer	Index Eligible Debt Out (\$bn)	% of IG Index	IG Index Rank	Composite Rating
ORCL	117	1.2%	5	BBB2
AMZN	117	1.2%	6	AA3
META	81	0.8%	12	AA3
GOOGL	74	0.7%	14	AA2
MSFT	34	0.3%	54	AAA
SPCX	25	0.2%	96	BBB1
Hyperscaler Total	448	4.4%		
AVGO	54	0.5%	23	A3
NVDA	33	0.3%	60	AA2
Total	535	5.3%		

Source: ICE, Morgan Stanley Research; Note: Sorted by Index debt outstanding.

Exhibit 14: AI-related exposure in credit is growing, but much lower than in equities



Source: Bloomberg, ICE BAML, Morgan Stanley Research

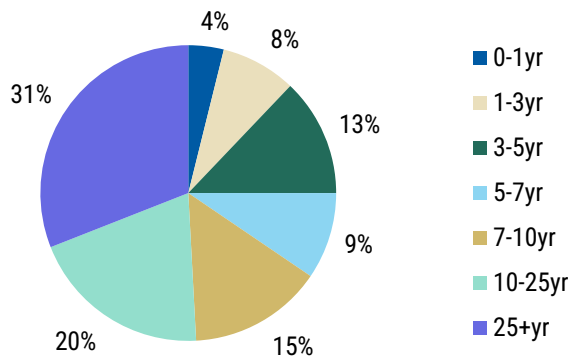
Exhibit 16: ORCL and AMZN are the largest AI-related names in the index

Rank	Ticker	Sector	Index Debt	% of Index
1	JPM	Banking	179	1.8%
2	MS	Financial Services	169	1.7%
3	BAC	Banking	166	1.6%
4	GS	Financial Services	139	1.4%
5	ORCL	Technology & Electronics	117	1.2%
6	AMZN	Retail	117	1.2%
7	WFC	Banking	106	1.0%
8	C	Banking	105	1.0%
9	T	Telecommunications	101	1.0%
10	HSBC	Banking	90	0.9%
11	VZ	Telecommunications	84	0.8%
12	META	Media	81	0.8%
13	CMCSA	Media	74	0.7%
14	GOOGL	Media	74	0.7%
15	UNH	Healthcare	70	0.7%

Source: ICE, Morgan Stanley Research; Note: Sorted by Index debt outstanding. Index-eligible debt rebalanced monthly and may not include intramonth issuance.

Exhibit 17: More than 50% of hyperscaler debt matures beyond 10 years

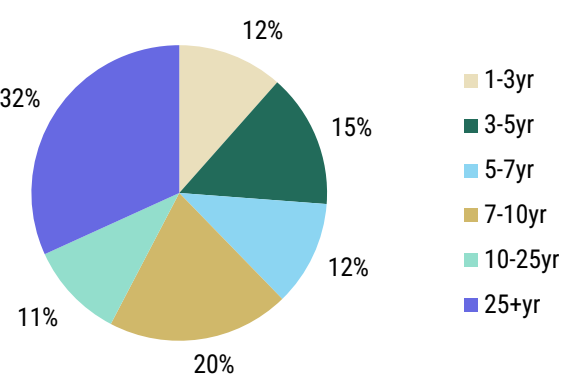
Hyperscaler Debt Tenor Profile (\$bn)



Source: Bloomberg, Morgan Stanley Research

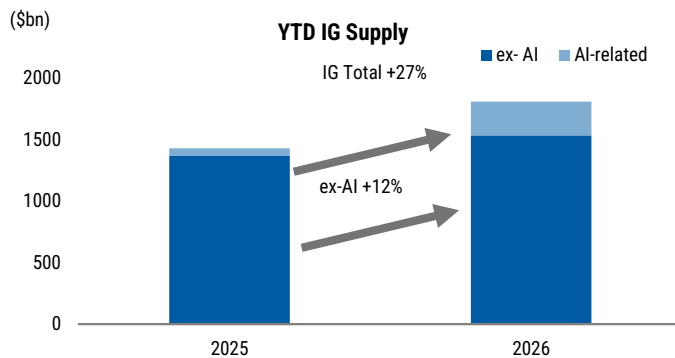
Exhibit 18: 2026 hyperscaler issuance is more concentrated in the 25+yr and 7-10yr bucket

2026 Hyperscaler Debt Tenor Profile (\$bn)



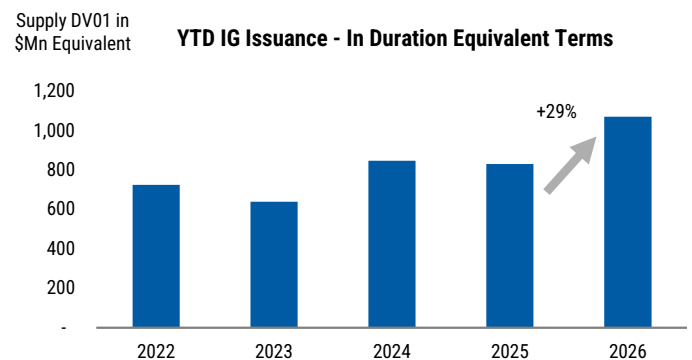
Source: Bloomberg, Morgan Stanley Research

Exhibit 19: The uptick in IG Supply YTD is largely driven by AI-related financing



Source: Dealogic, Morgan Stanley Research

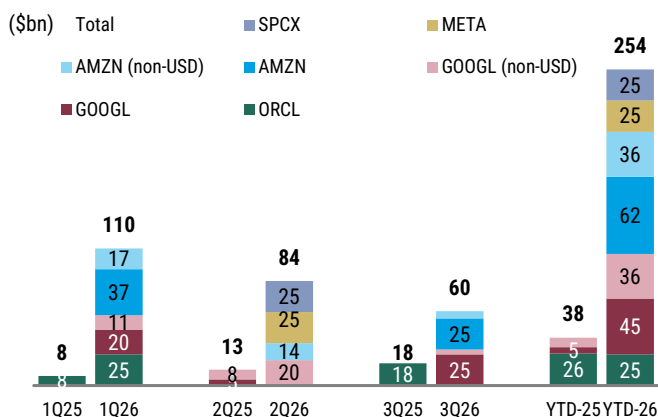
Exhibit 20: In duration terms, overall IG issuance is up 29% y/y, more on par with 2025 than earlier this year



Source: Dealogic, Morgan Stanley Research; Note: Issuance converted to duration terms in Mn (Notional *duration/100)

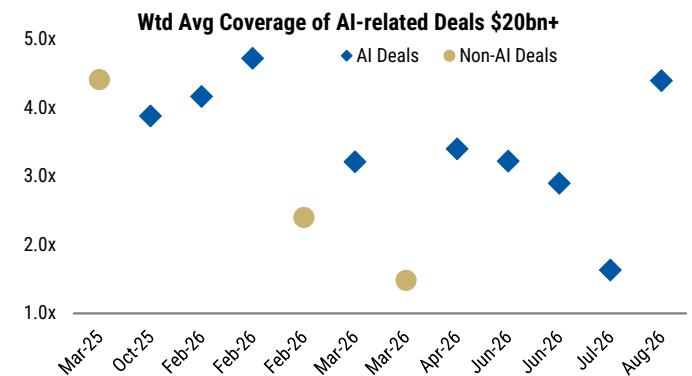
Hyperscalers Funding Mix & Balance Sheet Capacity

Exhibit 21: YTD issuance for the hyperscalers is at \$254bn (including SPCX), with ~\$72bn of non-USD debt



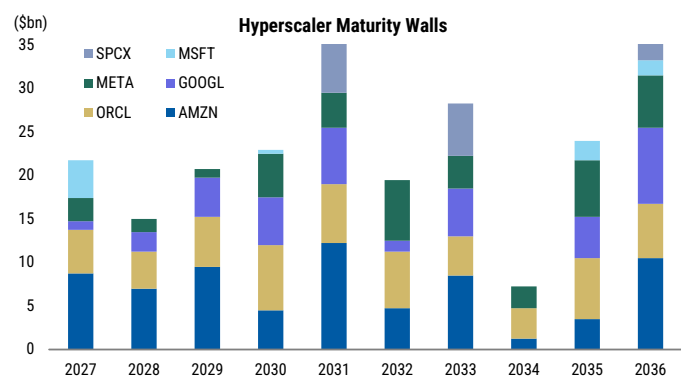
Source: Bloomberg, Morgan Stanley Research

Exhibit 22: Avg. across large AI-related deals since 4Q25 is 3.5x, but timing matters for individual deals



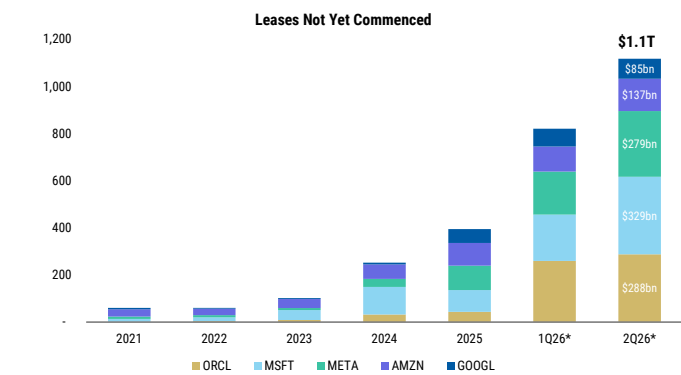
Source: Bloomberg, Morgan Stanley Research. Note: There have only been 3 non-AI deals of this size since 2025.

Exhibit 23: Hyperscalers have ~\$37bn in maturities over the next two years



Source: ICE BAML, Morgan Stanley Research.

Exhibit 25: Off-balance sheet leverage is rising as hyperscalers enter into long-dated leases for data center shells...



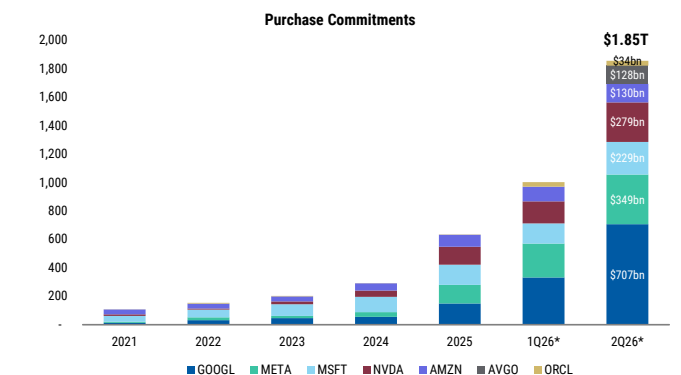
Source: Company Filings, Morgan Stanley Research. Note: Quarterly data for ORCL based on the two most recent filings.

Exhibit 24: Hyperscaler metrics more in line with broader Tech, but with double the cash-to-debt

Select Metrics by Sector 2Q26				
Sector	Gross Leverage	Net Leverage	Cash/Debt	Median Rating
Communication Services	1.7x	0.5x	64%	BBB+
Consumer	2.3x	1.7x	21%	BBB
Energy	1.7x	1.2x	16%	BBB
Health Care	2.6x	2.2x	22%	BBB+
Information Technology	1.7x	0.8x	52%	BBB
Hyperscalers	1.3x	0.4x	132%	AA-
Materials	2.3x	1.9x	15%	BBB
Utilities	5.8x	5.6x	1%	BBB
Non-Fin IG	2.2x	1.7x	19%	BBB

Source: FactSet, Bloomberg, Morgan Stanley Research. Note: For this exhibit, we're not yet including SPCX into the hyperscalers line. Leverage numbers based on reported debt and does not include off-balance sheet commitments. For off-balance sheet commitments please see [Exhibit 25](#)

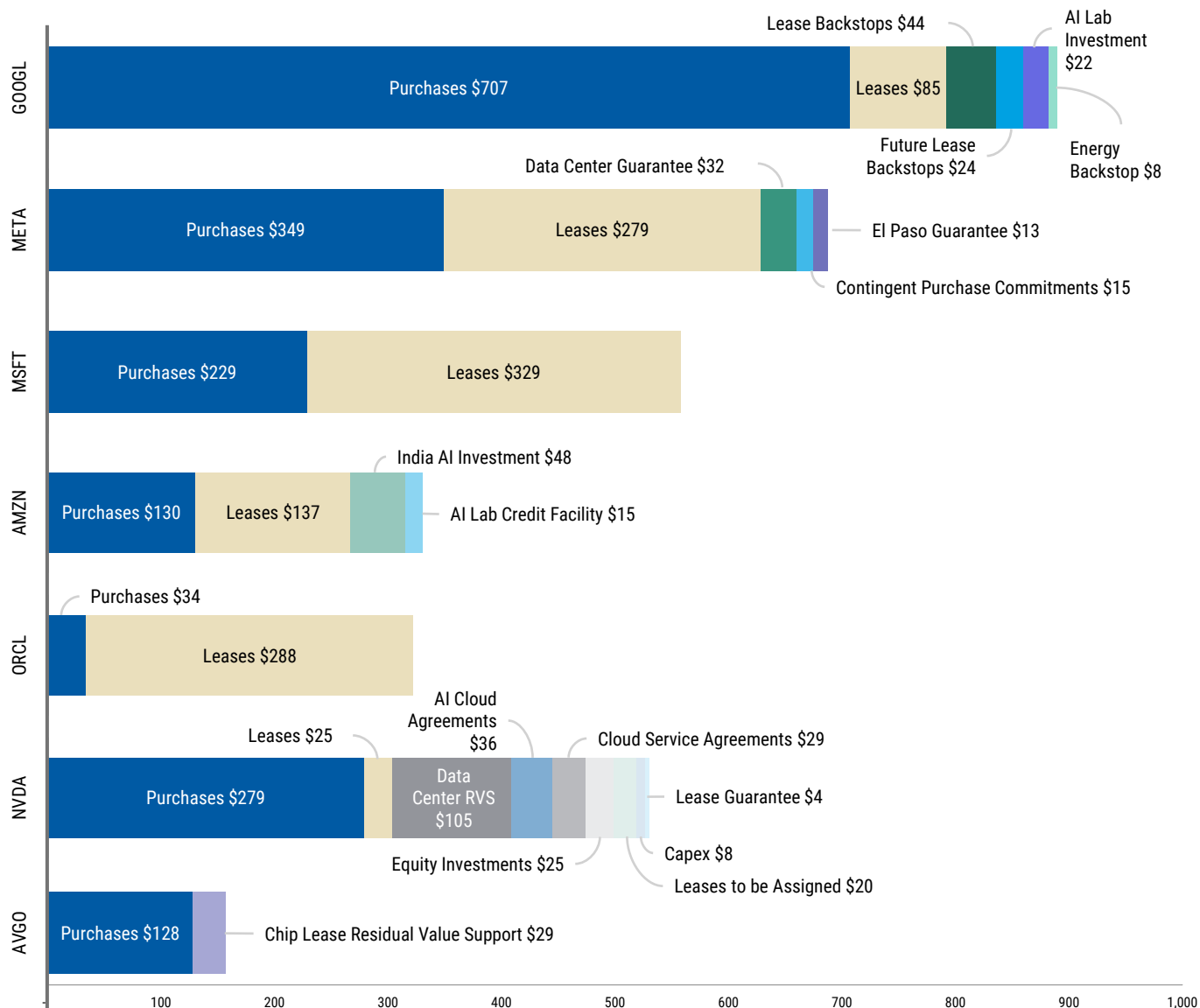
Exhibit 26: ...and commit to larger and longer-term purchases of compute, chips, equipment, and power



Source: Company Filings, Morgan Stanley Research. Note: Quarterly data for ORCL based on the two most recent filings.

Exhibit 27:

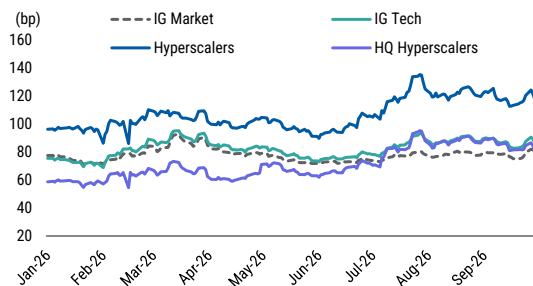
Hyperscalers, NVDA, AVGO disclose more than \$3.4tn of off-BS commitments and guarantees



Source: Morgan Stanley Research, Company Filings and Statements.

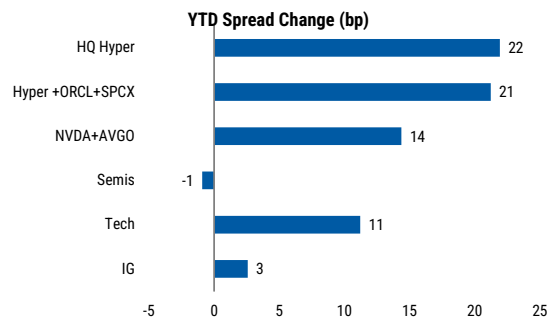
Hyperscaler Spread Performance

Exhibit 28: HQ Hyperscalers spreads are now just 1bp wider than the IG Index



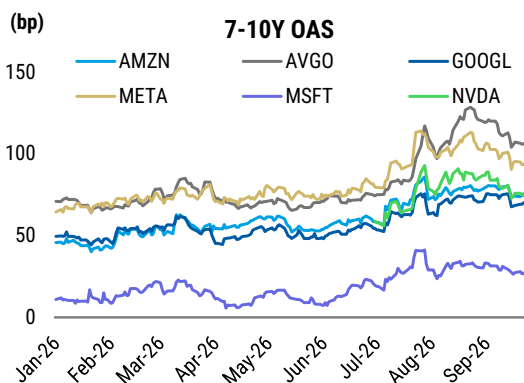
Source: Bloomberg, Morgan Stanley Research.

Exhibit 29: YTD widening most severe for the HQ Hyperscaler cohort, IG Index still only 3bps wider on the year



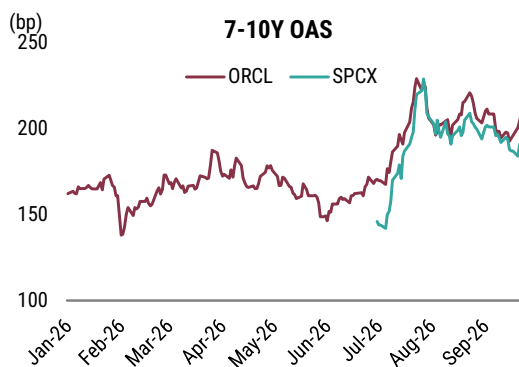
Source: Bloomberg, Morgan Stanley Research

Exhibit 30: AVGO 7-10Y bonds now trading wider to META



Source: ICE, Morgan Stanley Research

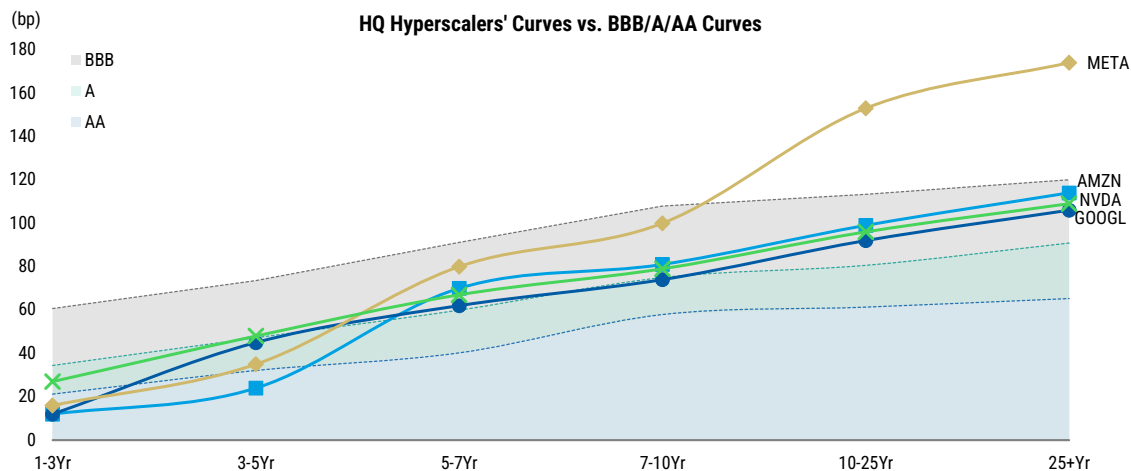
Exhibit 31: BBB AI-related names trade >100bp wider to the HQ cohort



Source: ICE, Morgan Stanley Research

Exhibit 32:

META long-end bonds trading closest to par are up to 54bp wider than the BBB curve



Source: ICE, Morgan Stanley Research. Data as of 9/25/2026. Note: The Ratings Curve excludes Financials, Utilities and HQ Hyperscalers (+ORCL SPCX & NVDA), Issuer spreads are derived from bonds closest to par within each tenor bucket.

Exhibit 33: IG AI-Related Tracker - Unsecured Corporate Spreads Tracker

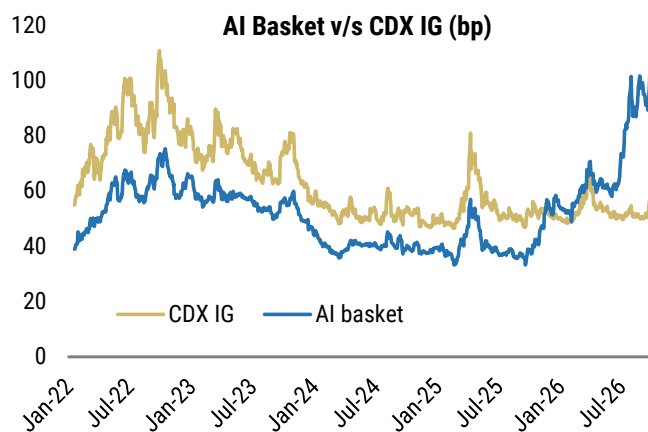
Security Name	Issue Date	Ratings (S/M/F)			Amt Out (\$mn)	Current Level	Spreads as of									Spread Change (bp)			
							8/31/26	7/31/26	6/30/26	5/29/26	4/30/26	3/31/26	2/27/26	1/30/26	1/5/26	MTD	QTD	Since 1Q26	YTD
MSFT 3.45 08/08/36	8/8/2016	AAA	Aaa	WD	1,749	32	39	55	40	28	40	39	43	36	35	(7)	(8)	(7)	(3)
MSFT 2.921 03/17/52	3/17/2021	AAA	Aaa	WD	6,250	70	76	79	71	58	61	68	69	58	56	(6)	(1)	+2	+14
NVDA 4 1/2 06/15/31	6/18/2026	AA	Aa1	NR	4,000	50	58	60	39	-	-	-	-	-	-	(8)	+11	-	-
NVDA 4.95 06/15/36	6/18/2026	AA	Aa1	NR	4,000	73	88	86	63	-	-	-	-	-	-	(15)	+10	-	-
NVDA 5 5/8 06/15/56	6/18/2026	AA	Aa1	NR	3,500	105	118	108	76	-	-	-	-	-	-	(13)	+30	-	-
GOOGL 4.1 02/15/31	2/13/2026	AA+	Aa2	NR	3,000	42	47	48	36	29	37	38	45	-	-	(5)	+6	+4	-
GOOGL 4.7 11/15/35	11/6/2025	AA+	Aa2	NR	3,500	68	71	67	56	50	59	60	61	53	55	(3)	+12	+8	+13
GOOGL 5.45 11/15/55	11/6/2025	AA+	Aa2	NR	4,000	103	108	104	84	72	80	79	86	74	76	(5)	+19	+24	+28
AMZN 4 1/4 03/13/31	3/13/2026	AA	A1	AA-	5,000	52	54	58	41	39	48	53	-	-	-	(2)	+11	(1)	-
AMZN 4.65 11/20/35	11/20/2025	AA	A1	AA-	3,500	73	73	78	61	51	65	64	64	54	56	(0)	+12	+9	+17
AMZN 5.45 11/20/55	11/20/2025	AA	A1	AA-	3,000	112	113	112	93	80	88	85	94	75	79	(1)	+19	+27	+33
META 4.55 05/15/31	5/4/2026	AA-	Aa3	NR	3,000	61	63	73	47	50	-	-	-	-	-	(2)	+14	-	-
META 4 7/8 11/15/35	11/3/2025	AA-	Aa3	NR	6,500	94	105	105	84	75	84	80	81	78	76	(11)	+10	+14	+19
META 5 5/8 11/15/55	11/3/2025	AA-	Aa3	NR	6,500	164	172	170	141	126	127	117	114	111	110	(8)	+23	+47	+54
AVGO 2.45 02/15/31	10/17/2025	A-	A3	A-	2,746	73	86	82	57	54	55	67	66	57	64	(13)	+16	+6	+9
AVGO 4.8 02/15/36	9/29/2025	A-	A3	A-	2,250	109	122	114	77	72	74	87	83	74	79	(13)	+32	+22	+30
AVGO 5.7 01/15/56	1/13/2026	A-	A3	A-	1,250	118	124	129	85	74	77	82	88	76	-	(6)	+33	+36	-
SPCX 5.35 07/15/31	6/26/2026	BBB	Baa1	BBB+	7,000	146	143	158	121	-	-	-	-	-	-	+3	+25	-	-
SPCX 5 7/8 07/15/36	6/26/2026	BBB	Baa1	BBB+	6,000	189	199	209	163	-	-	-	-	-	-	(10)	+26	-	-
SPCX 6.65 07/15/56	6/26/2026	BBB	Baa1	BBB+	3,500	226	239	243	201	-	-	-	-	-	-	(13)	+26	-	-
ORCL 4.95 02/04/31	2/4/2026	BBB-	Baa2	BBB	3,500	170	161	169	127	129	149	155	147	-	-	+9	+43	+15	-
ORCL 5.2 09/26/35	9/26/2025	BBB-	Baa2	BBB	4,000	210	204	206	170	152	180	177	173	161	162	+6	+40	+33	+48
ORCL 5.95 09/26/55	9/26/2025	BBB-	Baa2	BBB	3,500	263	252	252	229	205	234	240	232	200	203	+11	+34	+23	+60
5s10s Curves						Current	8/31/26	7/31/26	6/30/26	5/29/26	4/30/26	3/31/26	2/27/26	1/30/26	1/5/26	MTD	QTD	Since 1Q26	YTD
NVDA (4.95 '36 - 4.5 '31)						23	30	26	24	-	-	-	-	-	-	(7)	(1)	-	-
GOOGL (4.7 '35 - 4.1 '31)						25	24	19	20	21	22	22	16	-	-	+1	+5	+3	-
AMZN (4.65 '35 - 4.25 '31)						21	19	20	20	12	17	11	-	-	-	+2	+1	+10	-
META (4.875 '35 - 4.55 '31)						34	42	32	37	25	-	-	-	-	-	(8)	(3)	-	-
AVGO (4.8 '36 - 2.45 '31)						36	36	32	20	18	19	20	17	17	15	+0	+16	+16	+21
SPCX (5.875 '36 - 5.35 '31)						43	56	51	43	-	-	-	-	-	-	(13)	+0	-	-
ORCL (5.2 '35 - 4.95 '31)						40	43	37	43	23	31	22	26	-	-	(3)	(3)	+18	-
10s30s Curves						Current	8/31/26	7/31/26	6/30/26	5/29/26	4/30/26	3/31/26	2/27/26	1/30/26	1/5/26	MTD	QTD	Since 1Q26	YTD
MSFT (2.921 '52 - 3.45 '36)						38	37	24	31	30	21	29	26	22	21	+1	+7	+9	+17
NVDA (5.625 '56 - 4.95 '36)						33	30	22	13	-	-	-	-	-	-	+3	+20	-	-
GOOGL (5.45 '55 - 4.7 '35)						36	37	37	28	22	21	19	25	22	21	(1)	+8	+17	+15
AMZN (5.45 '55 - 4.65 '35)						40	40	34	32	29	23	21	30	21	24	(0)	+8	+19	+16
META (5.625 '55 - 4.875 '35)						70	67	65	57	51	43	37	33	33	35	+3	+13	+33	+35
AVGO (5.7 '56 - 4.8 '36)						8	2	15	8	2	3	-5	5	2	-	+6	+0	+13	-
SPCX (6.65 '56 - 5.875 '36)						38	40	34	38	-	-	-	-	-	-	(2)	+0	-	-
ORCL (5.95 '55 - 5.2 '35)						53	48	46	59	53	54	63	59	39	41	+5	(6)	(10)	+12

Source: Bloomberg, Morgan Stanley Research. Levels reflect bid spread to benchmark and as of 9/30/26.

Hyperscaler CDS Performance

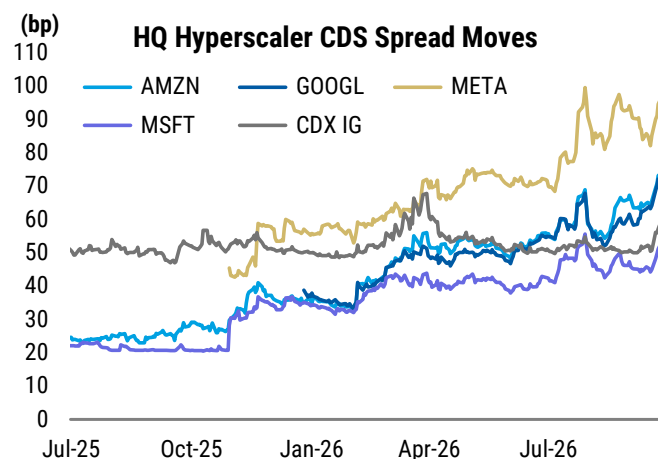
Buying CDS protection remains our preferred way of playing the AI story in the credit derivatives market. Liquidity across hyperscaler CDS have improved meaningfully since their inclusion in the CDX (March roll), and given the increasing AI-related risk concentration in a small set of names, we expect increasing needs for non-economic hedging to manage counterparty exposure (even in a benign scenario in which AI investments are profitable and therefore investments continue). Key risks to this trade: lower issuance vs. estimates, as well as faster-than-expected monetization results in stronger fundamentals, could result in CDS spreads tightening back.

Exhibit 34: CDS AI basket is now trading ~45bp wider to CDX IG



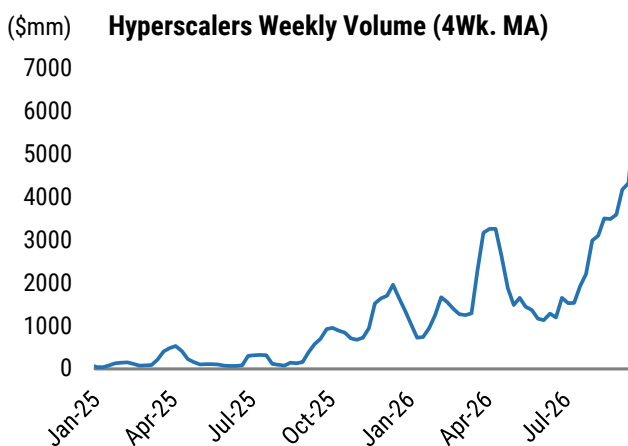
Source: S&P, Bloomberg, Morgan Stanley Research

Exhibit 36: MSFT 5Y CDS is the only one that still trades tighter to CDX IG



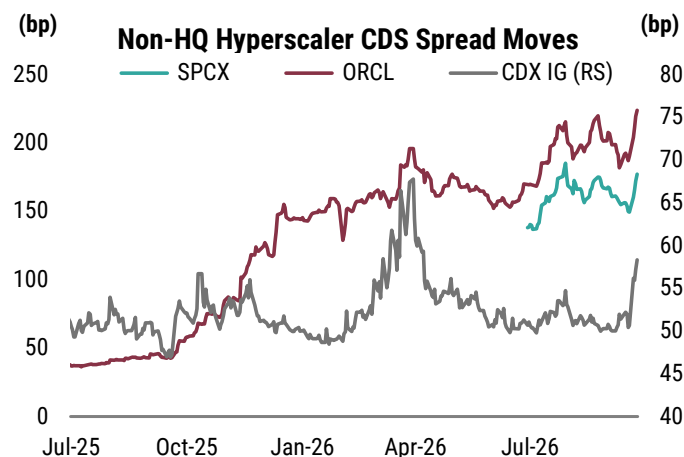
Source: S&P, Bloomberg, Morgan Stanley Research

Exhibit 35: Hyperscaler weekly volumes continue to increase



Source: DTCC, Morgan Stanley Research

Exhibit 37: ORCL 5Y CDS vs. SPCX 5Y CDS



Source: S&P, Bloomberg, Morgan Stanley Research

Collateral-Backed Risk in The IG Market

We reiterate our preference for collateral-backed risk over unsecured corporate risk.

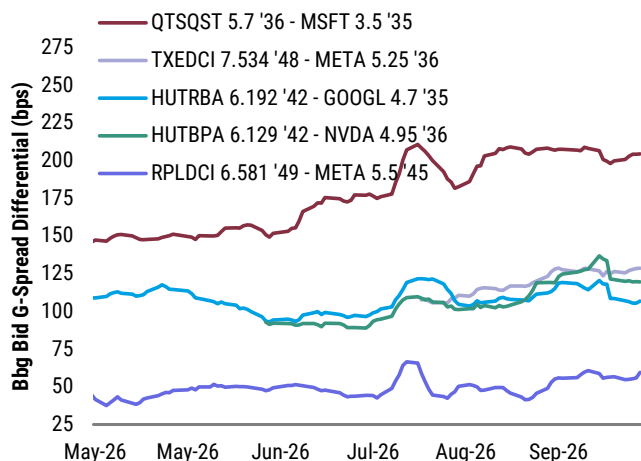
For data center-backed paper, this is driven by our view that, in addition to the long-term demand-supply imbalance for compute, near-term asset value might also benefit from incremental challenges to start new construction and secure power (please refer to our thematic strategists' work on the 3 Ps: people, **power**, and **politics**). Once construction is completed, we expect the differential between data center paper and tenants' unsecured bonds to compress. **We favor deals with high-quality tenants, residual value guarantees, full amortization through the life of the notes/lease, and documents that 1) require rent commencement once shells are delivered (regardless of power availability), and 2) offer adequate protections or bondholder participation in case of a lease reassignment.**

Exhibit 38: There's ~\$105bn of collateral-backed debt in IG public/private markets

Issue	Amt. Out. (\$bn)	Market	Index Inclusion	Ratings (S/M/F)	Spred to Bench (bps)
Data Center Bonds					
RPLDCI 6.581 05/30/49	27	144A Priv. Plac. ¹	-	A+ NR NR	254 220
QTSQST 5.7 04/15/36	5	144A Priv. Plac.	Bbg Global Agg ²	NR Baa2 NR	138 229
RDMICH 7 1/2 03/30/45	14	144A Priv. Plac.	-	NR Baa3 BBB	329 343
HUTRBA 6.192 11/15/42	3	144A Priv. Plac.	Bbg Global Agg ²	BBB- NR BBB-	185 184
HUTBPA 6.129 11/30/42	4	144A Priv. Plac.	Bbg Global Agg ²	NR Baa2 NR	165 200
TXEDCI 7.534 11/30/48	13	144A Priv. Plac. ¹	Bbg Global Agg ²	A+ NR AA-	288 239
QTSODY 6 5/8 09/01/31	4	144A Priv. Plac.	Bbg Global Agg ²	NR Baa3 BBB-	286 206
Total DC Financing	70				
Chip Financing - AVGO Transaction (4a2)					
Senior 'A1' Notes	6	Priv. Plac. (4a2)			100
Senior 'A2' Notes	25	Priv. Plac. (4a2)			
APCHIP 5 3/4 07/20/31	1	144A			165 185
APCHIP 5 3/4 10/20/35	23	Priv. Plac. (4a2)			
Loan format	1	Priv. Plac. (4a2)			
Junior 'B' Notes	4	Priv. Plac. (4a2)			440
Total Chip Financing	35				

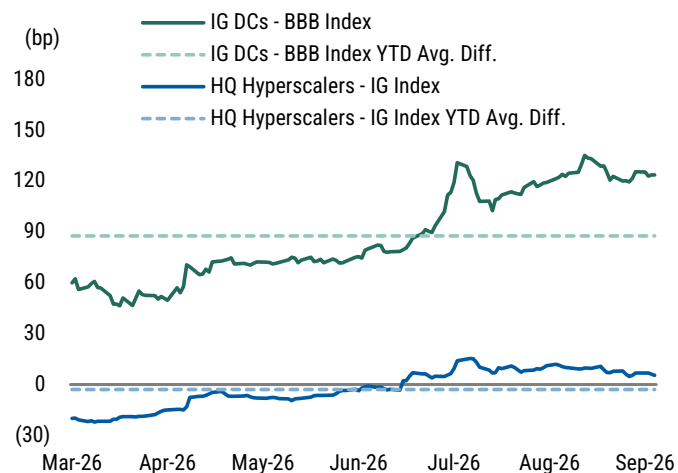
Source: Bloomberg, Morgan Stanley Corporate Trading Desk, Morgan Stanley Research. Notes: For chip financing, these are committed amounts. 1. Denotes the Beignet and Sopaiilla transactions. 2. These bonds are included in other indices as well. Levels as of 9/29/26.

Exhibit 40: IG DC bonds trade 50-200bp back of their tenants' unsecd bonds



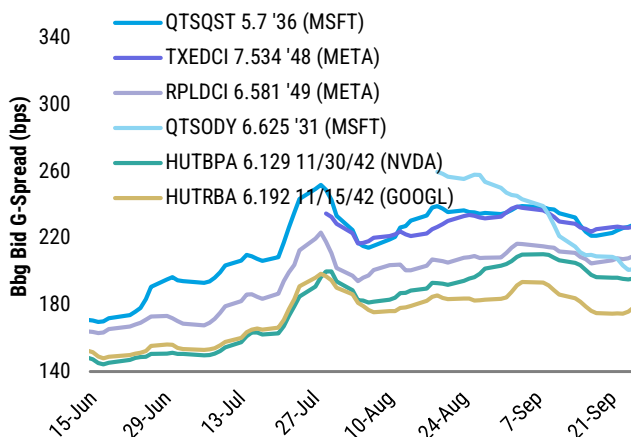
Source: Bloomberg, Morgan Stanley Research

Exhibit 39: IG DC bonds trade ~120bp wide to the BBB index



Source: Bloomberg, Morgan Stanley Research

Exhibit 41: IG DC bonds performance since June



Source: Bloomberg, Morgan Stanley Research

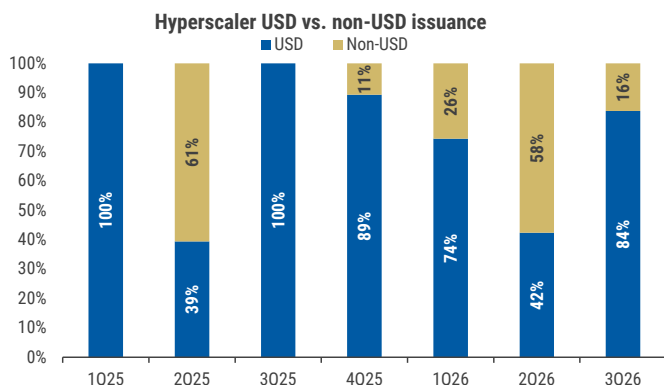
Investment Grade Reverse Yankee: Hyperscalers Continue to Tap Non-USD Markets

Hyperscalers Diversified Their Funding Base And Tapped Numerous Currencies This Year

While September saw a lull in AI-debt supply overall, non-USD hyperscaler issuance has continued to broaden. **This month saw a £4.25bn four-part GBP deal from AMZN, only a few weeks after GOOGL's debut A\$5.5bn six-parter in August.** The recent deals bring total non-USD hyperscaler supply to \$72bn: nearly a third of global hyperscaler supply this year. We maintain our expectations for further non-USD issuance from hyperscalers, with [recent media reporting](#) also pointing to potential EUR deals in the pipeline for 4Q. In contrast to 2025 when hyperscalers only tapped USD and EUR markets, less than 40% of non-USD supply this year has come in EUR. So far, issuers have diversified their funding towards five other currencies, including CAD, GBP, CHF, AUD, and JPY ([Exhibit 43](#)).

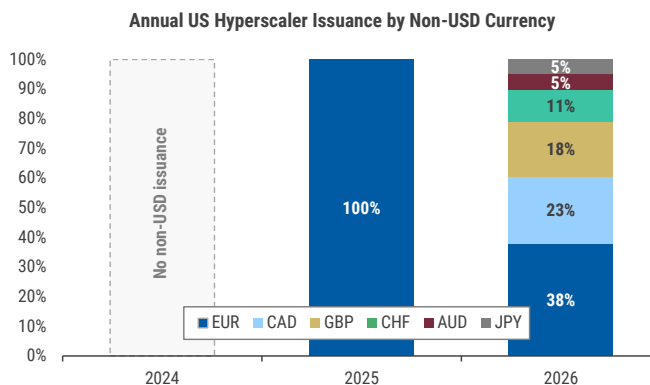
Given their relatively large deal sizes ([Exhibit 45](#) and [Exhibit 46](#)), these issuers have already become some of the largest issuers in many of the local IG benchmarks ([Exhibit 44](#)). However, since only two issuers (GOOGL and AMZN) have utilised the non-USD markets so far this year, sector concentration remains less pronounced in other markets, particularly EUR, AUD and GBP.

Exhibit 42: 2Q26 saw a majority of HQ hyperscaler supply coming in non-USD currencies



Source: Bloomberg, Bond Radar, Morgan Stanley Research

Exhibit 43: The pace of non-USD issuance from hyperscalers has proven resilient



Source: Bloomberg, Bond Radar, Morgan Stanley Research

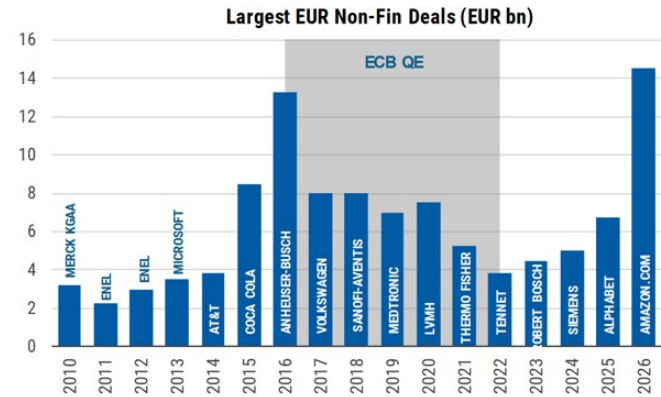
Exhibit 44:

Hyperscalers are already heavily represented in non-USD IG credit indices

Ticker Rating			Weight in IG Non-Fin Index							Rank in IG Non-Fin Index						
			USD	CAD	EUR	GBP	CHF	JPY	AUD	USD	CAD	EUR	GBP	CHF	JPY	AUD
Hyperscalers	AMZN	AA3	1.6%	3.4%	0.6%	2.0%	4.3%	0.0%	0.0%	1	5	30	6	5	-	-
	ORCL	BBB2	1.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2	-	-	-	-	-	-
	META	AA3	1.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	5	-	-	-	-	-	-
	GOOGL	AA2	1.0%	2.1%	1.0%	2.4%	4.7%	8.0%	3.0%	6	13	8	2	4	5	4
	MSFT	AAA	0.4%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	68	-	249	-	-	-	-
	Total		5.6%	5.4%	1.8%	4.5%	9.0%	8.0%	3.0%							
AI-adj.	AVGO	A3	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	14	-	-	-	-	-	-
	NVDA	AA3	0.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	43	-	-	-	-	-	-

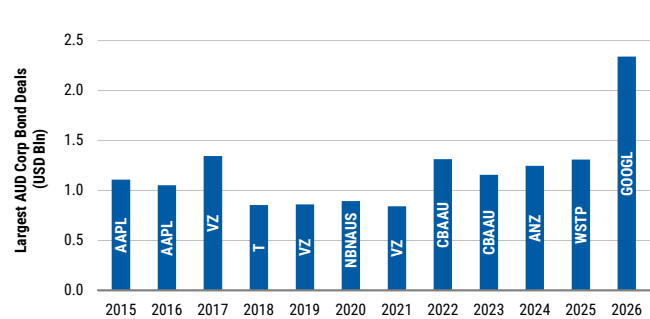
Source: Bloomberg, Bond Radar, Morgan Stanley Research; Note: Includes hyperscaler deals from August that are index-eligible but have not yet entered the latest index

Exhibit 45: Hyperscalers have shown EUR credit's ability to absorb QE-era-sized jumbo deals



Source: ICE BofA, Morgan Stanley Research

Exhibit 46: GOOGL's AUD deal was the largest bond deal on record in the AUD corp bond market



Source: Bloomberg, Morgan Stanley Research

Reverse Yankee Spread Performance

The performance of hyperscaler new issues this year has been mixed, with many deals seeing spread widening in the first 20 days of trading ([Exhibit 47](#)). Since July, hyperscaler new issue performance has been relatively more benign, mirroring the better relative performance of these issuers' secondary bonds. Even so, hyperscaler EUR bonds still trade wide to the AA benchmark, with AMZN trading closer to the A-rated index ([Exhibit 48](#)). The underperformance remains particularly pronounced in the long-end, with these issuers' curves screening steep compared to the overall market ([Exhibit 49](#)).

In the APAC LC space, GOOGL's AUD bonds spreads were tighter post issuance in August, outperforming the AUD LC corp bond market ([Exhibit 51](#)). In contrast, GOOGL's JPY bonds saw spread widening in the past two months, significantly underperforming the JPY LC corp bond market ([Exhibit 52](#)).

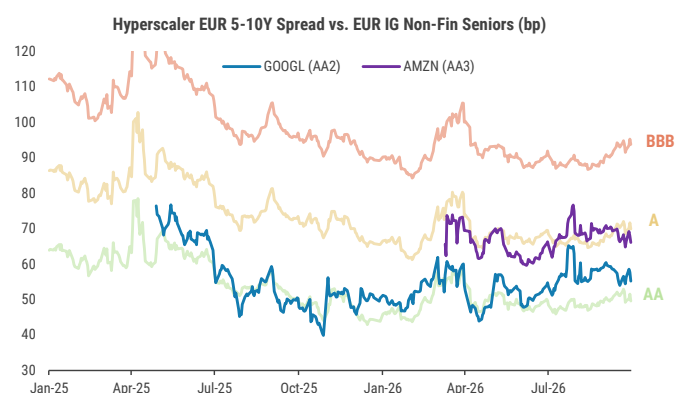
On a cross-currency adjusted spread basis, EUR hyperscaler bonds still trade modestly tight to their USD bonds, though the differential has narrowed since late August ([Exhibit 50](#)).

Exhibit 47: Hyperscaler new issues' performance has been mixed the month following pricing...

Avg. New Issue Spread (bp)				Cuml. Spread Change After Issue (days)																				
Date	Ccy	Ticker	Spread	NIC	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Feb-26	USD	ORCL	147	-21	-4	-1	+7	+11	+8	+12	+12	+13	+15	-	+18	+15	+13	+14	+23	+25	+20	+25	+37	+36
Feb-26	USD	GOOGL	63	9	-1	-3	-3	-2	-	-2	-3	-3	-4	-2	-2	-3	-2	+1	-1	-2	-4	-2	-1	-1
Feb-26	CHF	GOOGL	70	-	+1	+2	+2	+2	+0	+1	+1	+0	+1	+2	+1	+2	+2	+2	+1	+4	-0	+10	+1	-
Feb-26	GBP	GOOGL	67	-	-1	-1	+1	+2	+4	+1	-0	-2	-1	-1	-1	+1	+3	+6	+8	+1	+1	+6	+7	+4
Feb-26	EUR	NTT	93	0	+5	+9	+2	+11	+10	+2	+6	+12	+3	+2	+4	+4	+11	+9	+1	+6	-5	+16	+8	+6
Feb-26	GBP	NTT	85	-	+1	+4	+5	+8	+3	+1	+4	+7	+3	+4	+6	+7	+8	+9	+8	+12	+13	+10	+10	+10
Mar-26	USD	AMZN	79	15	-2	-3	-2	-3	-5	-5	-6	-7	-8	-10	-11	-10	-7	-7	-10	-14	-14	-16	-13	-
Mar-26	EUR	AMZN	93	-	+2	+4	+1	+0	+3	+3	+8	+5	+0	-0	+0	+2	+1	-2	-2	-	-	-1	-5	-
May-26	EUR	GOOGL	91	8	+0	+1	+4	+3	+2	+0	+0	-2	-1	-10	-1	-4	-2	-1	-1	+0	-1	-2	-7	-
May-26	CAD	GOOGL	74	-7	-5	-4	-3	-3	-4	-3	-4	-4	-	-4	-6	-5	-6	-6	-6	-7	-6	-7	-7	-8
May-26	CHF	AMZN	89	-	+1	-	-0	+3	+1	+1	+2	+2	-	+2	+2	+2	+2	+3	+3	+2	+3	+3	+2	-
May-26	JPY	GOOGL	79	-0	-2	+2	-0	+1	+2	+2	+2	+4	+4	+1	+5	+3	+4	+0	+1	+4	+3	+3	+3	+2
Jun-26	CAD	AMZN	70	-	-0	-0	-1	-1	-2	-1	-1	+1	+2	+3	+4	+4	+4	+5	+4	+4	-	+3	+3	-
Jun-26	USD	DELL	76	-7	-1	-1	+1	+2	+3	-	+3	+5	+7	+8	+10	+8	+9	+8	+6	-	+6	+10	+8	+8
Jun-26	USD	NVDA	43	9	+2	+3	+4	-	+4	+4	+6	+7	+7	+6	+6	+6	+5	-	+5	+11	+10	+12	+15	+14
Jun-26	USD	SPCX	144	-	+9	+14	+20	+17	+16	+18	+17	-	+15	+24	+24	+30	+42	+45	+52	+46	+44	+46	+49	+54
Jun-26	USD	NTT	83	1	-1	-1	-0	+0	-1	-2	-	-2	-3	-3	-4	-4	-5	-4	-4	-4	-4	-4	-4	-4
Jun-26	EUR	NTT	91	13	-	-5	-3	-4	-1	-2	-5	-7	-8	-10	-7	-7	-10	-4	-7	-4	-8	-6	-8	-5
Jun-26	GBP	NTT	80	-	-3	-3	-3	-3	-4	-4	-3	-4	-3	-2	-3	-3	-3	-2	-1	-2	-2	-2	-2	-
Jul-26	USD	AMZN	83	21	-5	-2	-1	-0	+2	-2	-4	-2	-3	-1	-1	+3	+7	+7	+11	+10	+5	+1	-1	-4
Jul-26	EUR	T	121	14	+0	+0	+1	-4	-5	-6	-6	-7	-6	-6	-5	-6	-7	-7	-7	-5	-4	-5	-5	-5
Aug-26	USD	GOOGL	81	21	-5	-3	-2	-6	-7	-5	-4	-3	-4	-4	-3	-4	-4	-5	-6	-7	-7	-5	-5	-5
Aug-26	AUD	GOOGL	117	-	-4	-1	-2	+1	-7	-6	-7	-8	-11	-5	-7	-10	-11	-10	-13	-19	-7	-8	-6	-3
Sep-26	GBP	AMZN	78	-	-3	-4	-4	-5	-6	-8	-8	-9	-8	-8	-7	-8	-6	-	-	-	-	-	-	-

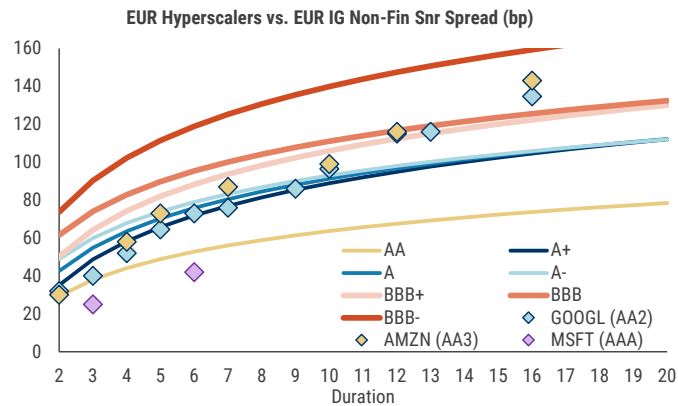
Source: Bloomberg, Bond Radar, Morgan Stanley Research

Exhibit 48: ...and in secondary, the underperformance of hyperscalers has slowed versus EUR IG



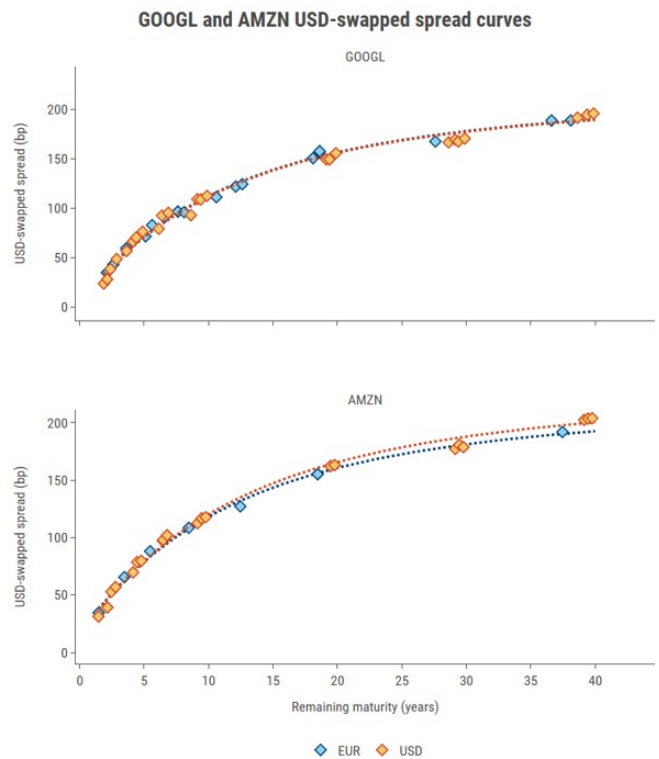
Source: Bloomberg, IHS Markit, Morgan Stanley Research; Note: Hyperscaler spreads reflect the simple average of mid Z-spread among bonds with maturities from 5-10 years

Exhibit 49: Hyperscaler credit curves are steep relative to their rating cohort



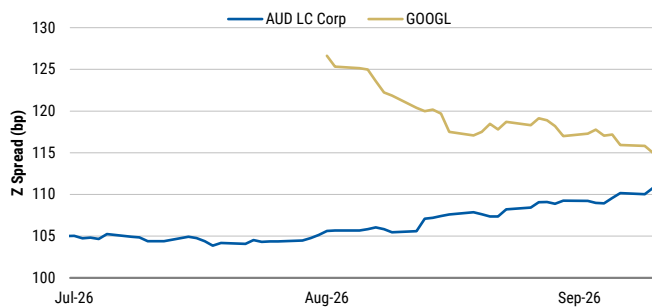
Source: ICE BofA, Morgan Stanley Research

Exhibit 50: On a cross-currency adjusted spread basis, hyperscaler EUR bonds trade modestly rich to their USD counterparts, though the gap has narrowed since late August



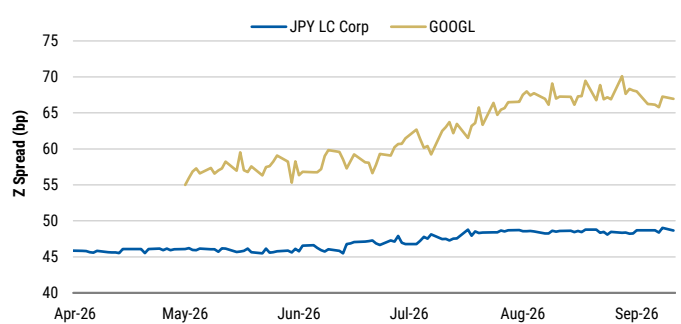
Source: Bloomberg, Bond Radar, Morgan Stanley Research; Note: Spreads reflect the cross-currency adjusted mid Z-spread at specific tenors, obtained by applying a Nelson-Siegel interpolation of existing USD/EUR-denominated, fixed-coupon bonds issued since 2025.

Exhibit 51: GOOGL's AUD spread has tightened while....



Source: Bloomberg, Morgan Stanley Research; Note: GOOGL's Z spread is weighted average Z spread of its AUD bonds

Exhibit 52: ...its JPY's spread has widened since issuance



Source: Bloomberg, Morgan Stanley Research; Note: GOOGL's Z spread is weighted average Z spread of its JPY bonds

We summarize relevant book stats and price/spread performance for hyperscaler issuance across European currencies below.

Exhibit 53: Selected reverse yankee deals – performance at a glance

Pricing Date	Ccy of Issue	Security	Amount Issued (local ccy mns)	Book Stats			Price				Spreads As Of (bp)			
				New Issue Premium (bp)	Bid-Cover (x)	IPT-Launch Spread Compression (bp)	01-Oct-26	Δ Since Issue	Δ YTD	Δ 1M	01-Oct-26	Δ Since Issue	Δ YTD	Δ 1M
29-Apr-25	EUR	GOOGL 2 1/2 05/06/29	1,500	-	4.2x	-33	97	-2.3	-2.2	-0.5	8	-44	-14	-11
		GOOGL 3 05/06/33	1,500	-	4.3x	-33	93	-6.0	-4.7	-1.4	60	-12	+8	-3
		GOOGL 3 3/8 05/06/37	1,250	-	4.2x	-30	91	-8.0	-5.8	-1.5	84	-11	+11	-6
		GOOGL 3 7/8 05/06/45	1,250	-	4.2x	-30	87	-13.2	-8.4	-0.7	132	+2	+31	-10
		GOOGL 4 05/06/54	1,250	-	4.6x	-30	83	-15.3	-8.6	-0.4	158	-2	+32	-6
03-Nov-25	EUR	GOOGL 2 3/8 11/06/28	1,000	-	2.8x	-35	98	-2.0	-1.8	-0.4	3	-22	-16	-8
		GOOGL 2 7/8 11/06/31	1,000	18	3.5x	-33	95	-4.7	-3.7	-1.1	40	-7	-3	-6
		GOOGL 3 1/8 11/06/34	1,000	8	3.4x	-33	92	-6.7	-4.9	-1.3	65	+3	+6	-7
		GOOGL 3 1/2 11/06/38	1,000	6	2.4x	-33	90	-9.0	-6.5	-1.6	96	+14	+17	-5
		GOOGL 4 11/06/44	1,250	18	3.1x	-30	88	-10.6	-8.3	-1.0	131	+16	+29	-7
		GOOGL 4 3/8 11/06/64	1,250	-	3.8x	-32	85	-13.3	-9.6	-0.3	186	+28	+40	-7
10-Feb-26	GBP	GOOGL 4 1/8 02/13/29	750	-	4.9x	-	98	-2.3	-	-0.5	33	-12	-	+3
		GOOGL 4 5/8 11/13/32	1,250	-	3.8x	-	95	-4.8	-	-1.1	47	-8	-	+3
		GOOGL 5 1/2 11/13/41	1,250	-	3.5x	-	90	-8.6	-	-1.0	74	+19	-	-1
		GOOGL 5 7/8 02/13/58	1,250	-	5.7x	-	87	-11.1	-	-0.5	85	+25	-	-1
		GOOGL 6 1/8 02/13/2126	1,000	-	9.8x	-	89	-10.4	-	+0.9	145	+25	-	+2
10-Feb-26	CHF	GOOGL 0.427 03/01/29	905	-	-	-10	99	-0.9	-	-0.5	35	-5	-	+0
		GOOGL 0.89 03/03/32	700	-	-	-5	98	-1.8	-	-0.8	56	-4	-	-5
		GOOGL 1.2525 03/03/36	575	-	-	-5	97	-2.9	-	-1.4	76	+6	-	+1
		GOOGL 1.5823 03/01/41	455	-	-	-3	95	-4.7	-	-1.7	102	+20	-	+4
		GOOGL 1.8675 03/03/51	420	-	-	-2	97	-3.3	-	-1.1	109	+11	-	+2
11-Mar-26	EUR	AMZN 2.8 03/16/28	1,250	-	1.8x	-25	99	-0.9	-	-0.3	4	-26	-	-6
		AMZN 3.1 03/16/30	2,000	-	2.1x	-27	98	-2.3	-	-1.0	33	-15	-	-2
		AMZN 3.35 03/16/32	2,250	-	2.1x	-27	96	-3.5	-	-1.3	57	-6	-	-4
		AMZN 3.7 03/16/35	2,500	-	2.3x	-27	95	-4.5	-	-1.4	78	-5	-	-6
		AMZN 4.05 03/16/39	2,250	-	1.6x	-27	94	-5.5	-	-1.7	101	-2	-	-5
		AMZN 4.45 03/16/45	1,250	-	2.0x	-28	93	-6.1	-	-1.3	134	-1	-	-6
		AMZN 4.85 03/16/64	1,250	-	2.1x	-25	92	-7.9	-	-0.9	192	+4	-	+0
05-May-26	EUR	GOOGL 3.2 05/11/30	1,500	7	1.8x	-33	98	-1.9	-	-0.8	28	-7	-	-9
		GOOGL 3.45 05/11/32	1,750	11	1.9x	-28	97	-2.9	-	-1.2	52	-3	-	-6
		GOOGL 3 5/8 05/11/34	1,500	4	2.0x	-32	96	-3.8	-	-1.4	65	+2	-	-6
		GOOGL 4.1 05/11/39	1,750	7	1.9x	-32	95	-5.0	-	-1.5	98	+5	-	-7
		GOOGL 4 1/2 05/11/45	1,250	16	2.7x	-30	94	-6.1	-	-1.3	136	+11	-	-5
		GOOGL 4.8 05/11/63	1,250	11	2.1x	-30	92	-7.6	-	-0.6	189	+14	-	-1
12-May-26	CHF	AMZN 0.835 05/28/29	900	-	-	+3	100	-0.2	-	-0.5	43	-12	-	-1
		AMZN 1.2 05/28/32	585	-	-	+3	99	-1.1	-	-1.1	72	-3	-	+1
		AMZN 1.4475 05/28/35	475	-	-	+3	98	-1.7	-	-1.5	87	+2	-	+3
		AMZN 1.675 05/28/38	390	-	-	+3	98	-2.4	-	-1.4	103	+8	-	+0
		AMZN 1.9389 05/25/44	245	-	-	+1	98	-2.3	-	-1.2	114	+6	-	+1
		AMZN 2.0825 05/25/51	225	-	-	+1	98	-2.0	-	-0.8	124	+6	-	+1
15-May-26	JPY	GOOGL 1.965 05/16/29	135,500	-	-	-3	99	-1.4	-	-0.2	48	+13	-	+3
		GOOGL 2.412 05/16/31	200,500	-	-	-3	98	-2.3	-	-0.5	64	+14	-	+8
		GOOGL 2.822 05/16/33	123,200	-	-	+1	98	-2.3	-	-0.5	66	-2	-	+4
		GOOGL 3.189 05/16/36	64,900	-	-	-3	96	-4.2	-	-0.2	81	+6	-	-5
		GOOGL 3.713 05/16/41	19,100	-	-	-1	95	-4.6	-	-1.8	87	-2	-	+10
		GOOGL 4.395 05/16/56	9,300	-	-	-1	96	-3.5	-	+1.9	101	-8	-	-14
19-Aug-26	AUD	GOOGL 4.599 05/17/66	24,000	-	-	+2	96	-4.3	-	+1.9	125	+0	-	-12
		GOOGL 5.2 08/27/29	500	-	3.4x	-	99	-0.8	-	-0.5	54	-11	-	-1
		GOOGL 5 1/2 08/27/31	750	-	3.7x	-	98	-1.5	-	-1.2	84	-6	-	+5
		GOOGL 6 1/4 08/27/36	1,000	-	3.5x	-	99	-0.7	-	-1.0	106	-27	-	-9
14-Sep-26	GBP	GOOGL 6.9 08/27/46	1,000	-	2.6x	-	97	-2.2	-	-2.2	167	-13	-	-1
		AMZN 5.2 09/14/29	1,250	-	2.5x	-	100	-0.2	-	-	41	-12	-	-
		AMZN 5.55 09/14/32	1,000	-	3.3x	-	99	-0.6	-	-	61	-14	-	-
		AMZN 6 1/4 09/14/38	1,000	-	2.3x	-	99	-0.9	-	-	84	-6	-	-
		AMZN 6.65 09/14/45	1,000	-	2.0x	-	99	-0.3	-	-	88	-5	-	-

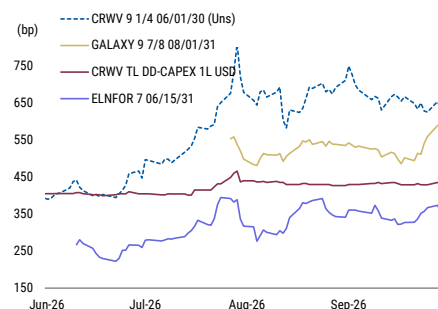
Source: Bloomberg, Bond Radar, Morgan Stanley Research estimates; Note: Bid Z-spread is shown for EUR, CHF, JPY bonds, bid G-spread for GBP bonds, ASW spread for AUD.

Leveraged Finance: Corporate Issuance Drove Supply Last Month

AI-related leveraged finance issuance continues to grow and is now at ~\$63bn, with HY issuance exceeding \$50bn YTD and leveraged loans reaching over \$10bn. Loan issuance slowed in September following a busier July and August, with just one \$509mn GPU financing reaching the market. Within HY, AI exposure rose from 3% to 4% of the HY index during September; we saw two data center-backed deals and one unsecured corporate deal in the form of a record \$10bn USD corporate HY offering, alongside \$1.14bn equivalent in EUR. Despite this jumbo unsecured issuance, AI-related debt in leveraged finance remains predominantly secured/project-finance-style (~70%).

AI-related issuance in HY is now at our full-year estimate of \$50bn, but we expect more supply in 4Q. Our [equity colleagues](#) expect more lease announcements throughout October, as this would be the latest possible date to begin construction on data centers expected to come online by year-end 2027. At the same time, we think project-level financing is more sensitive to higher interest rates (see our [more detailed analysis here](#)), and so unless lease economics are strong, the current environment might pose an additional temporary challenge to supply (on top of increasing political pushback and challenges related to power). Therefore, we expect 4Q supply to be more similar to 1Q26/3Q26 than to 2Q26 ([Exhibit 57](#)).

Exhibit 54: Collateral backed loans/bonds trade tighter than CRWV's unsecured bonds



Source: Bloomberg, Morningstar, Morgan Stanley Research. Spreads are OAS for bonds, DMargin for the loan.

Collateral-backed paper in the HY market trades inside of unsecured paper: there is a ratings differential given the benefit of asset value for recovery, and investors can now choose between chips and shells as collateral for certain HY issuers ([Exhibit 54](#)).

Given the different nature of the assets, chips offer a faster path to monetization vs. powered shells. We expect more clarity over the next few months on how credit support from higher-quality semiconductor names could play a part in accelerating chip financing supply in the HY and leveraged loan markets. With regard to data center bonds, we favor deals with high-quality tenants or IG credit support, lower LTC, and some deleveraging through amortization before refi, and documents that 1) require rent commencement once shells are delivered (regardless of power availability), and 2) offer adequate protections or bondholder participation in case of a lease reassignment. Headline risk around construction delays could continue to drive bouts of volatility.

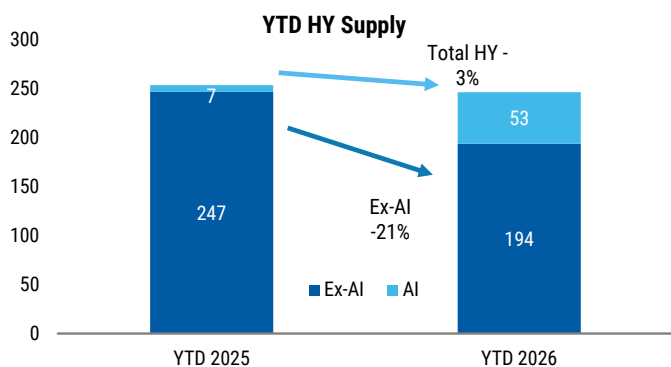
Similar to what we wrote earlier for IG, the HY market is also facing headwinds from the magnitude and velocity of the rate sell-off. HY yields are now at +8.3%, and total returns are approaching negative territory. Demand has been choppy, with HY seeing outflows in four out of the past five weeks. Macro uncertainty has played out alongside a heavy

issuance calendar, with total supply in September at ~\$50bn, including the two largest deals in the history of the market, which priced at the end of the month.

HY spreads in aggregate have widened to 300bp+, and the BBB-BB basis has gone from 50bp to 80bp. Notably, the widening in BBs has not been limited to data center-related issuance. Rather, it has happened broadly across the market, likely driven by both outflows and heavy issuance. In our view, the impact of higher rates matters most for CCCs as it exacerbates existing debt sustainability and refinancing worries, even if growth is good. CCCs have widened by 400bp this year and spreads are hitting at 1000bp. However, for high-quality HY, we think risk-reward looks good when rates stabilize, particularly as the issuance calendar gets absorbed by the market and earnings growth remains robust.

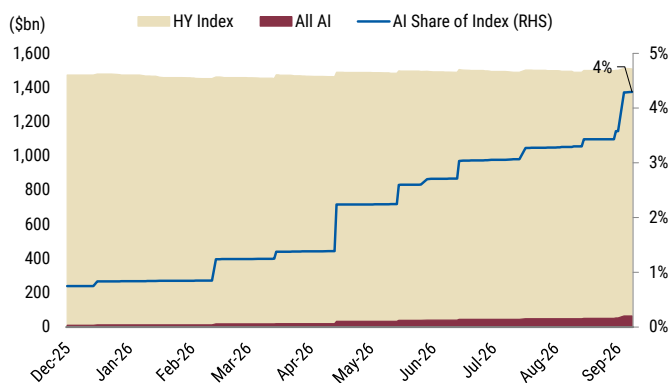
AI-Related Debt is ~4% of the HY Index; HY DCs Trade Wide to BB Index

Exhibit 55: Ex-AI, HY YTD supply is down 21% y/y



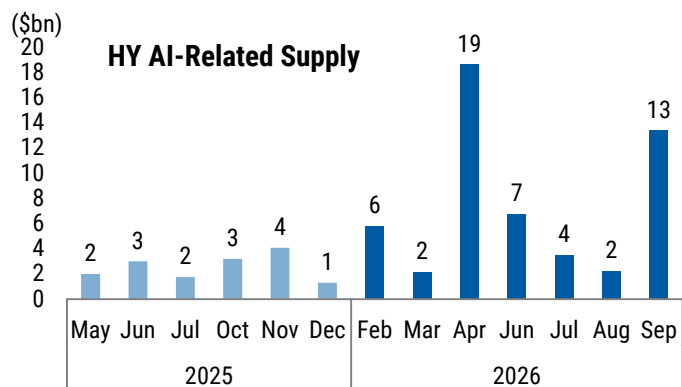
Source: Pitchbook LCD, Morgan Stanley Research; As of 9/28/2026

Exhibit 56: AI-related debt accounts for ~4% of the HY index, up from 3% in August



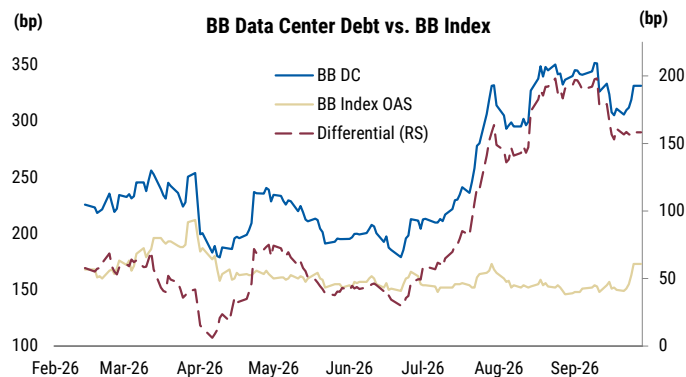
Source: ICE BofA, Morgan Stanley Research

Exhibit 57: After a slow start to the month, we saw the second-highest monthly AI-related supply in September at \$13bn.



Source: Pitchbook, Morgan Stanley Research

Exhibit 58: BB-rated HY data center-backed deals are now trading 160bp back from the BB index, down from 200bp at the month's start.



Source: Bloomberg, Morgan Stanley Research

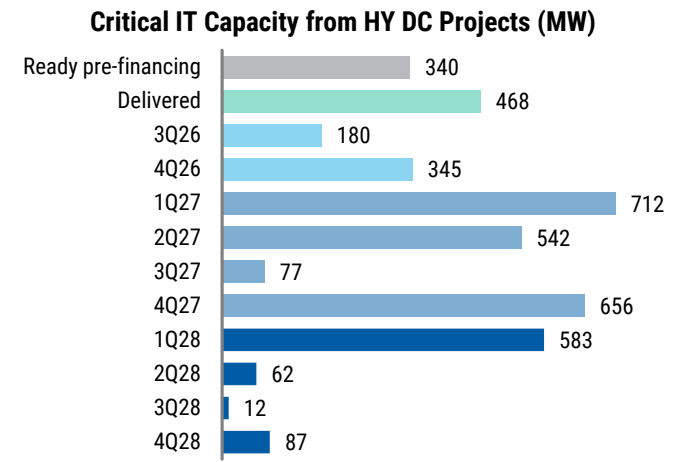
HY Data Center Bonds Tracker and Performance Trends

Exhibit 59: There are now 20 project finance-style bonds in the HY market spanning eight tenants.

Issue	Amt Out (\$bn)	Pricing Date	YTC (%)	S&P	Moody's	Fitch	Location	Critical IT MW	Tenant	Add'l Credit Support	First Call Date	Final Delivery Date
BLKPRL 6.125 '31	2.00	2/4/2026	9.9%	NR	Ba2	BB-	Wink, TX	216	Amazon	-	2/15/2028	2/1/2027
STNGRY 6.000 '31	0.81	6/8/2026	10.6%	NR	Ba2	BB-	Andrews, TX	70	Amazon	-	6/15/2028	4/1/2027
APLD 9.250 '30	2.35	11/13/2025	9.2%	BB-	NR	BB-	Ellendale, ND	250	CoreWeave	-	12/15/2027	10/31/2026
CORZ 7.750 '31	3.30	4/22/2026	12.4%	BB	NR	BB-	TX,GA,OK, NC	590	CoreWeave	-	5/15/2028	6/30/2027
EDGCOM 7.500 '31	1.30	4/21/2026	16.0%	BB+	NR	BB-	Chicago/Atlanta	114	CoreWeave/Alibaba	-	4/30/2028	12/1/2027
ELKGPV 7.500 '31	0.90	6/2/2026	13.3%	BB-	Ba3	BB-	Chicago, IL	72	CoreWeave	-	6/15/2028	1/1/2027
ELNFOR 7.000 '31	1.59	6/9/2026	13.1%	BB-	NR	BB-	Ellendale, ND	150	CoreWeave	-	6/15/2028	6/1/2027
GALAXY 9.875 '31	3.51	7/28/2026	15.1%	BB-	NR	BB-	Lubbock, TX	260	CoreWeave	-	8/1/2028	3/31/2028
DDRIVE 8.875 '31	1.12	9/23/2026	13.3%	BB-	NR	BB-(EXP)	Richmond, VA	76	CoreWeave	-	9/30/2028	3/31/2028
CIFR 7.125 '30	1.73	11/5/2025	9.6%	NR	Ba3	BB-	Colorado City, TX	207	Fluidstack	\$1.733bn Google backstop	11/15/2027	3/31/2027
FLASHC 7.250 '30	1.30	12/18/2025	10.6%	NR	Ba3	NR	Abernathy, TX	168	Fluidstack	\$1.3bn Google backstop	12/31/2027	9/30/2026
MERIDI 6.250 '31	5.70	4/16/2026	13.4%	BB-	Ba2	BB	Sullivan County, IN	430	Fluidstack	Google lease guarantee	4/30/2028	2/1/2028
WULF 7.750 '30	3.20	10/16/2025	9.0%	BB	Ba2	BB	Lake Mariner, NY	438	Fluidstack	\$3.2bn Google backstop	10/15/2027	12/31/2026
ZENARC 8.875 '31	2.25	8/13/2026	16.6%	BB	Ba2	NR	Oklahoma	149	Jane Street	-	8/31/2028	10/1/2028
CLSPRK 7.875 '31	2.28	9/18/2026	10.2%	NR	Ba2	BB-(EXP)	Sandersville, GA	175	Meta	-	10/1/2028	12/31/2027
TRACTC 5.875 '31	3.80	2/13/2026	16.2%	(P)BB+	Ba1	BB	Storey County, NV	200	NVIDIA	-	3/1/2028	6/1/2028
TRACTD 6.500 '31	4.59	4/28/2026	15.3%	(P)BB	Ba1	BB	Storey County, NV	200	NVIDIA	-	5/1/2028	4/4/2028
PFORGE 6.750 '31	2.15	3/3/2026	12.1%	NR	Ba3	BB-	Harwood, ND	200	Oracle	-	3/15/2028	6/1/2027
YNDRDC 6.875 '31	0.72	6/23/2026	13.6%	NR	Ba3	BB	Loudoun County, VA	48	Oracle	-	6/30/2028	10/1/2028
SECMOS 8.875 '31	1.00	4/30/2026	14.0%	(P)BB	NR	BB-	Austin, TX	50	SoftBank	-	5/1/2028	6/1/2027
Total:	45.59							4,063				

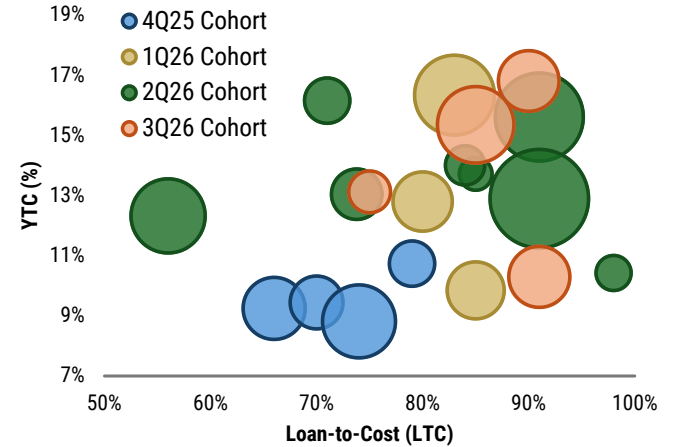
Source: Bloomberg, Morgan Stanley Research; Data as of 9/28/2026

Exhibit 60: Capacity delivery tracker based on public disclosures and updates



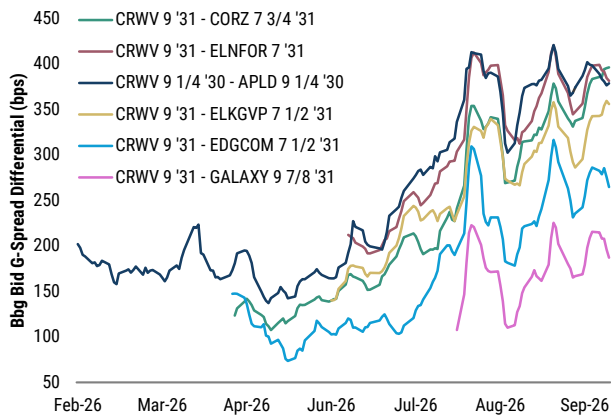
Source: Bloomberg, Moody's, Fitch, S&P, Morgan Stanley Research. Changes vs. last month: See CIFR's announcement for Barber Lake (we moved capacity to 1Q27), included September projects.

Exhibit 61: LTC vs. YTC: 2Q26 cohort remains the one with most dispersion



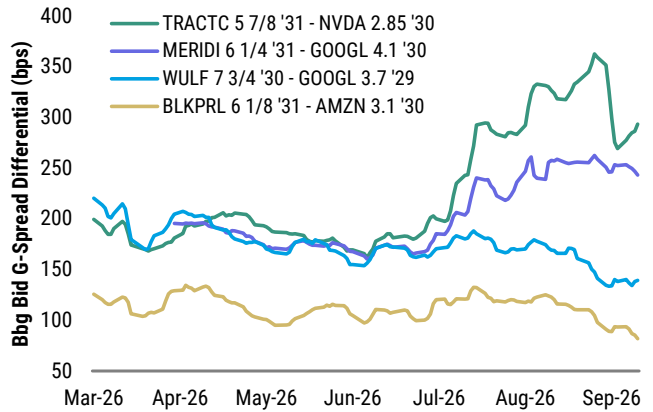
Source: Bloomberg, Moody's, Fitch, S&P, Morgan Stanley Research. Note: bubble size represents amount outstanding for each bond.

Exhibit 62: CRWV unsecured bonds vs. HY DC bonds that have CRWV as a tenant



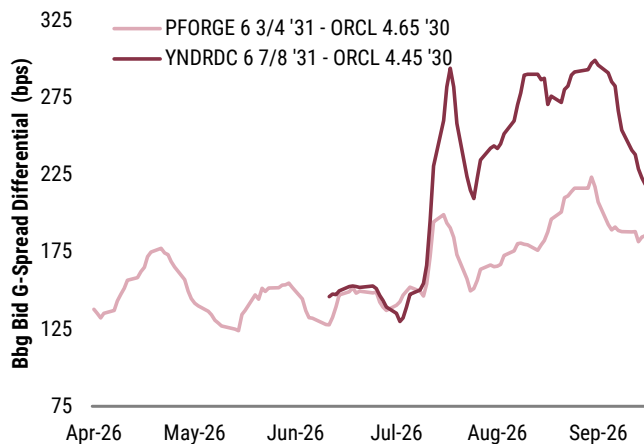
Source: Bloomberg, Morgan Stanley Research

Exhibit 63: HY DC bonds with IG tenants or credit support trade 80-240bp back of the tenant's unsecd bonds; TRACT/NVDA relationship now an exception



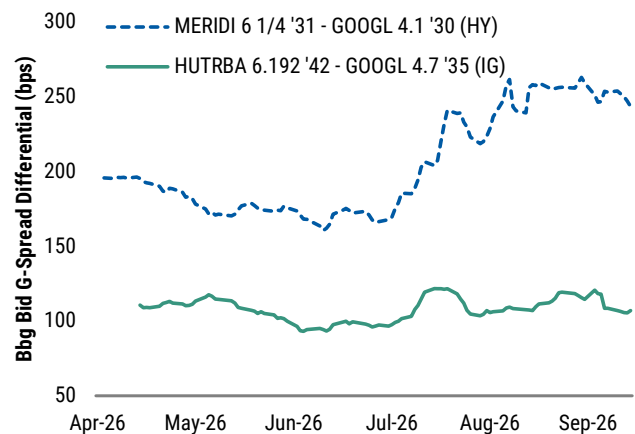
Source: Bloomberg, Morgan Stanley Research

Exhibit 64: HY DC bonds with ORCL as a tenant vs. ORCL unsecd bonds of similar duration



Source: Bloomberg, Morgan Stanley Research

Exhibit 65: IG vs. HY DC bonds: Wider differential for HY DC bonds driven by construction and refinancing risks



Source: Bloomberg, Morgan Stanley Research

Exhibit 66:

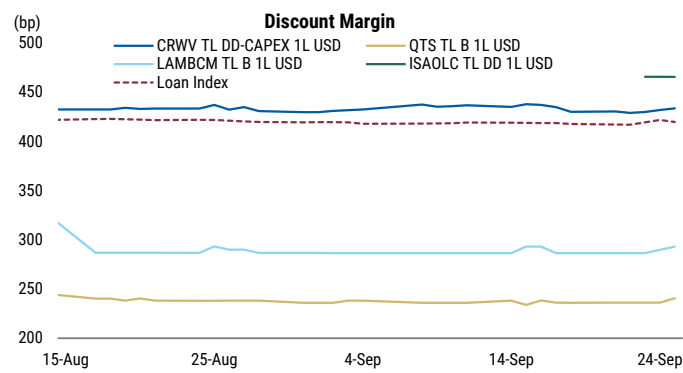
HY Issuer Performance – Debt & Equity

							Spread Change (Equity % Change)								
Type	Entity	Description	Seniority	Price	Yield	Spread	1d	1wk	2wk	1mth	Year to Date (since issuance)	S&P	Moody's	Fitch	Out (bm)
HY	BLACK PEARL COMPUTE LLC	BLKPRL 6 1/8 02/15/31	Secured	98.65	6.48	133	+14	+18	+14	-9	-117	-	Ba2	BB-	2.00
HY	STINGRAY COMPUTE	STNGRY 6 06/15/31	Secured	96.54	6.87	173	+11	+7	-5	-28	-4	-	Ba2	BB-	0.81
HY	APLD COMPUTECO LLC	APLD 9 1/4 12/15/30	Secured	104.75	7.77	218	+2	+7	+4	-37	-374	BB-	-	BB-	2.35
HY	EDGED COMPUTE LLC	EDGCOM 7 1/2 04/30/31	Secured	92.22	9.92	498	+14	+47	+48	-4	+160	BB+	-	BB-	1.30
HY	CORE SCIENTIFIC FINANCE	CORZ 7 3/4 05/15/31	Secured	97.09	8.56	342	+3	+3	-14	-53	-52	BB	-	BB-	3.30
HY	ELK GROVE VILLAGE PROPER	ELKGVP 7 1/2 06/15/31	Secured	94.95	8.98	394	+11	+14	+6	-15	+75	BB-	Ba3	BB-	0.90
HY	APLD COMPUTECO 3 LLC	ELNFOR 7 06/15/31	1st lien	93.82	8.73	367	+10	+40	+28	+3	+100	BB-	-	BB-	1.59
HY	GALAXY HELIOS DATA CNTR	GALAXY 9 7/8 08/01/31	Secured	96.99	10.68	559	+19	+58	+55	+26	+6	BB-	-	BB-	3.51
HY	DDC 01 PROPCO LLC	DDRIVE 8 7/8 09/30/31	1st lien	98.33	9.30	413	+15	-	-	-	+15	BB-	-	BB-(EXP)	1.12
HY	WULF COMPUTE LLC	WULF 7 3/4 10/15/30	Secured	102.94	6.67	142	+11	+21	+14	-21	-151	BB	Ba2	BB	3.20
HY	CIPHER COMPUTE LLC	CIFR 7 1/8 11/15/30	1st lien	101.04	6.83	143	-0	+3	-23	-60	-122	-	Ba3	BB-	1.73
HY	FLASH COMPUTE LLC	FLASHC 7 1/4 12/31/30	Secured	99.71	7.35	220	+6	+11	-14	-25	-135	-	Ba3	-	1.30
HY	MERIDIAN ARC HOLDCO LLC	MERIDI 6 1/4 04/30/31	Secured	93.83	7.87	277	+5	-7	+1	-10	+82	BB-	Ba2	BB	5.70
HY	ZENITH ARC LLC	ZENARC 8 7/8 08/31/31	Secured	91.70	11.09	599	+8	-36	+38	+76	+97	BB	Ba2	-	2.25
HY	SV RNO PROPERTY OWNER 1	TRACTC 5 7/8 03/01/31	Secured	91.33	8.32	355	+33	+52	-12	-8	+162	(P)BB+	Ba1	BB	3.80
HY	PR RNO PROPERTY OWNER 1	TRACTD 6 1/2 05/01/31	Secured	92.12	8.65	386	+31	+63	-8	-1	+143	(P)BB	Ba1	BB	4.59
HY	APLD COMPUTECO 2 LLC	PFORGE 6 3/4 03/15/31	1st lien	95.03	8.25	309	+6	+8	+7	-1	-3	-	Ba3	BB-	2.15
HY	YONDR JK 1 LLC	YNDRDC 6 7/8 06/30/31	1st lien	93.03	8.84	374	+2	+10	-42	-49	+143	-	Ba3	BB	0.72
HY	SE COSMOS LLC	SECMOS 8 7/8 05/01/31	1st lien	96.73	9.77	451	+1	+10	-16	-3	-8	(P)BB	-	BB-	1.00
HY	CSDC FINANCE I LLC	CLSPRK 7 7/8 10/01/31	1st lien	99.21	8.07	276	+7	-	-	-	-6	-	Ba2	BB-(EXP)	2.28
HY	COREWEAVE INC	CRWV 9 1/4 06/01/30	Sr Unsecured	93.50	11.45	625	-3	-42	-6	-55	-132	B	B1	BB-	2.00
HY	COREWEAVE INC	CRWV 9 02/01/31	Sr Unsecured	88.63	12.46	729	-7	-3	+44	+7	-21	B	B1	BB-	1.75
HY	COREWEAVE INC	CRWV 9 3/4 10/01/31	Sr Unsecured	87.69	13.18	808	-4	-6	+73	+30	+295	B	B1	BB-	2.75
HY	COREWEAVE INC	CRWV 9 5/8 07/15/32	Sr Unsecured	85.80	13.18	800	-3	+2	+65	+43	+296	B	B1	BB-	1.25
HY	IRON MOUNTAIN INC	IRM 6 1/4 01/15/35	Sr Unsecured	96.14	6.87	144	+11	+13	+18	+5	+1	BB-	Ba3	-	1.50
HY	SOFTBANK GROUP CORP	SOFTBK 9 3/4 04/01/34	Sr Unsecured	100.28	9.70	451	+11	-	-	-	+11	BB+	-	BB+	4.50
HY	SOFTBANK GROUP CORP	SOFTBK 8 5/8 04/01/30	Sr Unsecured	101.72	8.02	287	+3	-	-	-	+3	BB+	-	BB+	1.00
HY	SOFTBANK GROUP CORP	SOFTBK 9 1/4 04/01/32	Sr Unsecured	100.35	9.17	401	+7	-	-	-	+7	BB+	-	BB+	4.50
Equity	COREWEAVE INC	CRWV Equity	-	87.59	-	-	-2.8%	7.7%	-1.6%	-0.5%	22.3%	-	-	-	-
Equity	GALAXY DIGITAL INC	GLXY Equity	-	24.24	-	-	-3.8%	-1.8%	-0.7%	-2.2%	8.4%	-	-	-	-
Equity	TERAWULF INC	WULF Equity	-	15.74	-	-	-3.4%	-7.5%	-6.0%	-3.6%	37.0%	-	-	-	-
Equity	APPLIED DIGITAL CORP	APLD Equity	-	26.25	-	-	-3.0%	-6.7%	-0.6%	-9.0%	7.1%	-	-	-	-
Equity	CIPHER DIGITAL INC	CIFR Equity	-	17.73	-	-	-2.0%	-3.3%	5.2%	8.7%	20.1%	-	-	-	-
Equity	SOFTBANK GROUP CORP	SOFTBK Equity	-	6150 JPY	-	-	-3.2%	-2.6%	-6.0%	20.9%	39.8%	-	-	-	-
Wavg BB OAS 341															
Total Outstanding															64.84

Source: Bloomberg, Morgan Stanley Research; As of 9/28/2026. For bonds issued in 2026, YTD performance is measured since issuance.

Leveraged Loans: Loan Supply Skews to Chip Financing

Exhibit 67: CRWV's May GPU financing loan now trades within 10bp of the broader index.



Source: Bloomberg, Pitchbook LCD, Morgan Stanley Research

Exhibit 68: We've seen five public loan transactions for AI-related assets; four for chip financing

Pricing Date	Issuer	Size (\$bn)	UoP	Cost	Offtake	Ratings	Maturity Date
5/15/2026	CRWC	3.1	GPU	SOFR+450	AI Lab / Cohere Inc	Ba2/BB+	11/17/2031
7/24/2026	QTS	3.3	Data Center	SOFR+225	QTS	Baa3/BBB	7/25/2033
8/7/2026	CRWC	2.6	GPU	SOFR+550	Jane Street / Hudson River Trading	Ba2/BB+	9/30/2031
8/10/2026	LAMBCM	0.9	GPU	SOFR+300	NVIDIA	Baa1	12/31/2030
9/23/2026	IS AN03	0.5	GPU	SOFR+425	-	Ba1/BB	11/29/2030

Source: Bloomberg, Morgan Stanley Research

Securitized Credit: Issuance into a Higher Rate Environment

After a relatively quiet summer, issuance is starting to pick up again in securitized credit, albeit slowly. Two ABS deals and one CMBS deal priced in September (\$1.5bn in ABS and \$375mm in CMBS). This comes as the 5y Treasury crossed 5%, leaving cost of debt more than 100bps higher than at the start of the year for data center ABS.

Spreads are weaker coming out of the summer months even as the pace of supply remains slightly low, moving in sympathy at last with the broader weakness in AI and data-center related financings in other pockets of the market. We continue to think risks skew to the downside for securitized credit data centers, particularly given our expectation that the market will continue to need to absorb the elevated supply through the end of the decade.

Two notable structural developments in the securitized credit space last month are also worth highlighting: in ABS, the first floating-rate ABS tranche was issued on September 30, a departure from the otherwise fixed-only asset class. In CMBS, partial amortization was proposed in an upcoming deal. Moreover, Vantage priced the lowest-LTV AAA tranche that we have seen thus far in ABS, at ~25%. We think these modifications reflect both the sponsor's need to crowd in more demand to contain spreads and alter the structure to accommodate the higher rate environment.

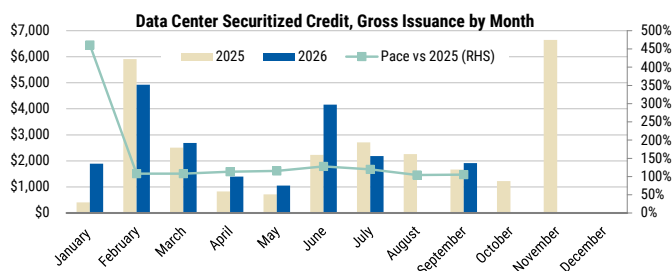
We recently modified our original supply forecasts to account for the higher rate environment, while taking into account expectations for significant new leased capacity additions through 2029. We brought our gross issuance expectation for 2026 down to \$25-30bn with ~\$17bn in ABS and ~\$10bn in CMBS (down from \$30-35bn). For 2027, we lowered our forecast to \$40-50bn, down from our previous estimate of ~\$55bn. For 2028, we hold our forecast steady at \$55-65bn. In total, that means we expect \$120-\$145bn of gross issuance across ABS and CMBS through 2028 (inclusive of 2026).

See our latest note for more thoughts on how the higher rate environment affects supply and the financing of stabilized assets:

[*Securitized Products & US Credit Strategy: Data Center Financing: Funding Stabilized Assets in a Higher Rate Environment \(24 Sep 2026\)*](#)

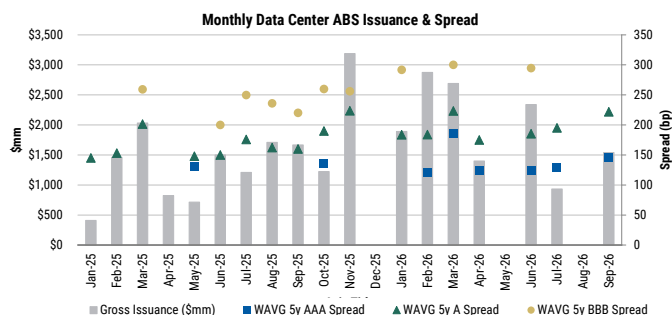
Data Center Securitized Credit Issuance

Exhibit 69: Data center issuance is now pacing in-line with 2025, with \$1.9bn of new deals in September



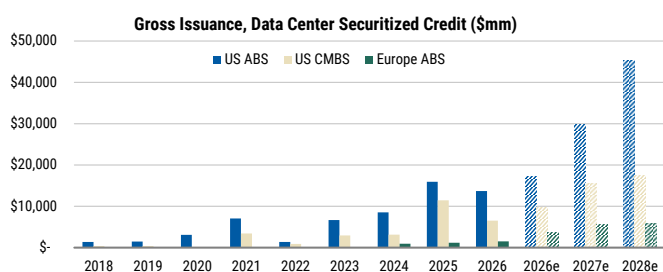
Source: CreditFlow, Intex, Morgan Stanley Research. Data through Sept 24, 2026

Exhibit 71: The cadence of issuance has been fairly steady in ABS this year, though spreads are now leaking wider



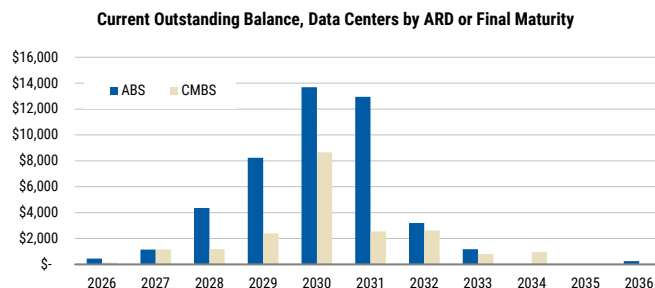
Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research

Exhibit 73: Gross issuance & forecast, data center securitized credit



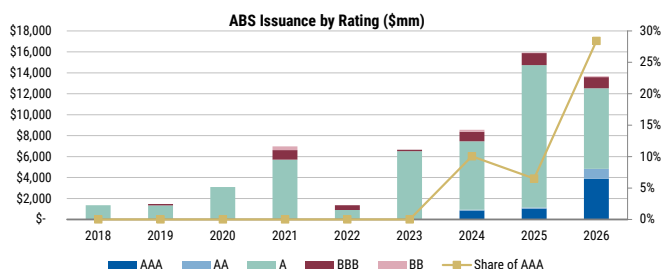
Source: CreditFlow, Trepp, Intex, Morgan Stanley Research

Exhibit 70: While incremental issuance continues to add to the 2030-2031 maturity wall that has been building for some time



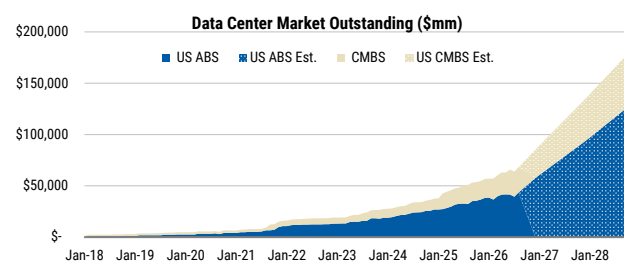
Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research. Data through Sept 24, 2026

Exhibit 72: Increasingly issuers are bringing AAA-rated paper to the market in an effort to crowd in more demand



Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research

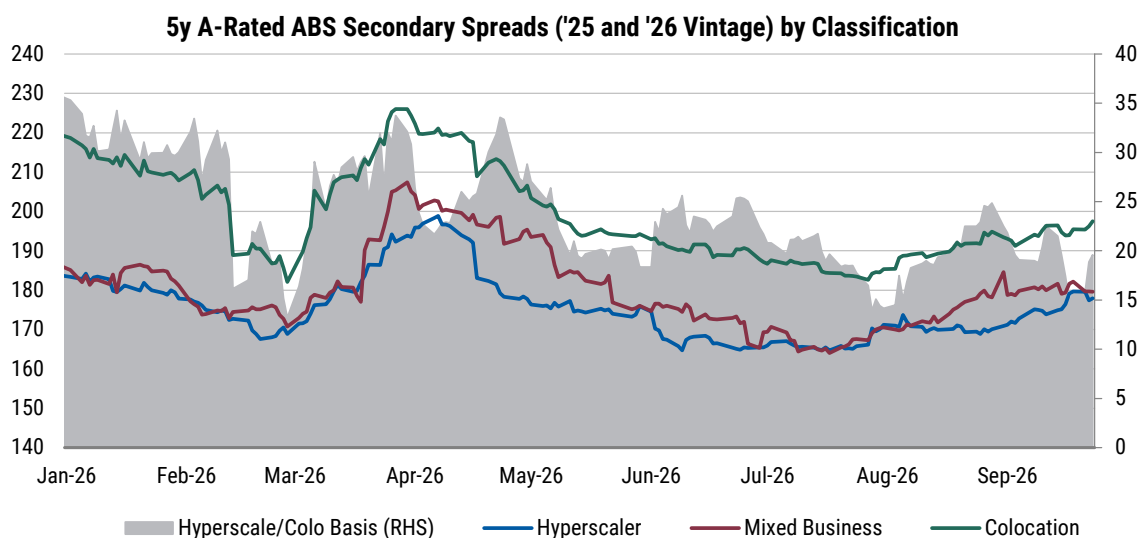
Exhibit 74: Outstanding balance & forecast, data center securitized credit



Source: CreditFlow, Trepp, Intex, Morgan Stanley Research

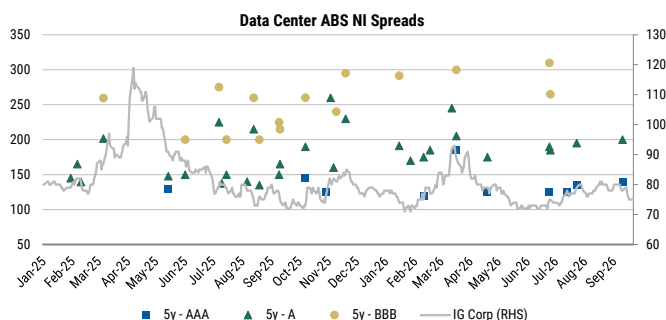
Data Center ABS Spread Performance

Exhibit 75: The basis between colocation & hyperscaler spreads moves largely in tandem with the direction of spreads, tightening throughout the course of the year. We prefer colocation over hyperscaler shelves.



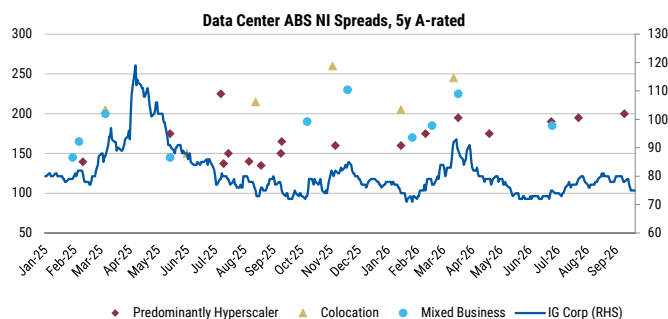
Source: Bloomberg, Morgan Stanley Research

Exhibit 76: New issue data center ABS Spreads (5y WAL)



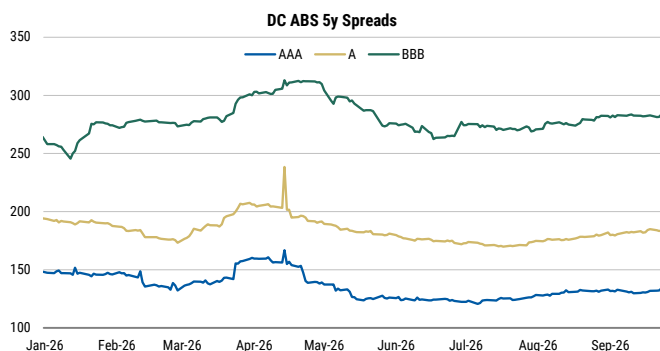
Source: CreditFlow, Morgan Stanley Research. Data through Sep 25th.

Exhibit 77: New issue data center spreads, 5y WAL, A-rating, by classification



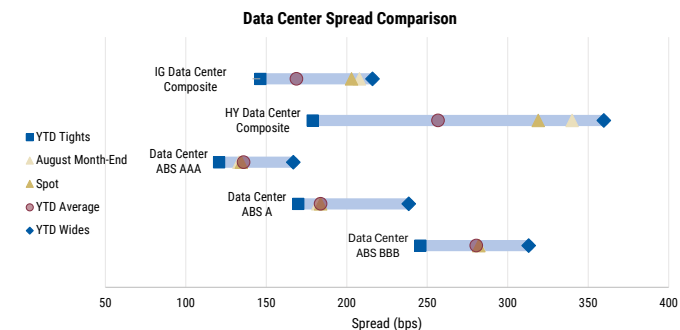
Source: CreditFlow, Morgan Stanley Research. Data through Sep 25th.

Exhibit 78: Secondary spreads, data center ABS, 5y WAL, by rating



Source: Bloomberg, Morgan Stanley Research. Includes only 2025 and 2026 vintage. Data through Sep 25th.

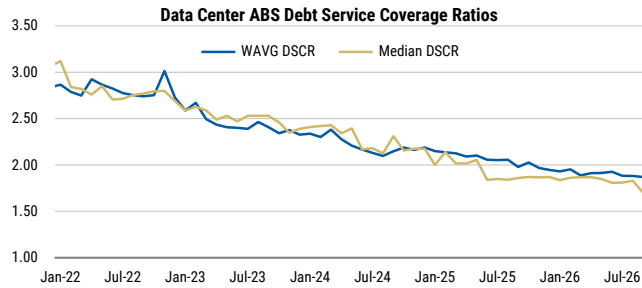
Exhibit 79: Data center ABS spreads vs secured data center corporate credit spreads



Source: Bloomberg, Morgan Stanley Research. Data through Sep 25th.

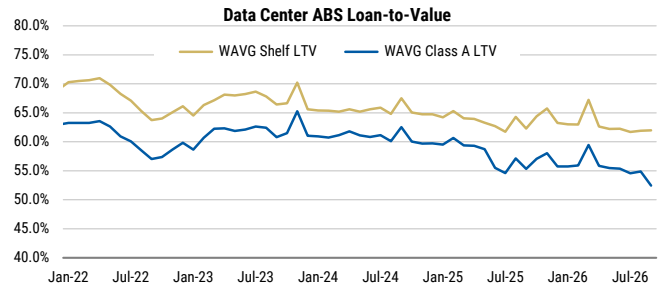
Data Center ABS Performance Metrics

Exhibit 80: Data center ABS Debt Service Coverage Ratios



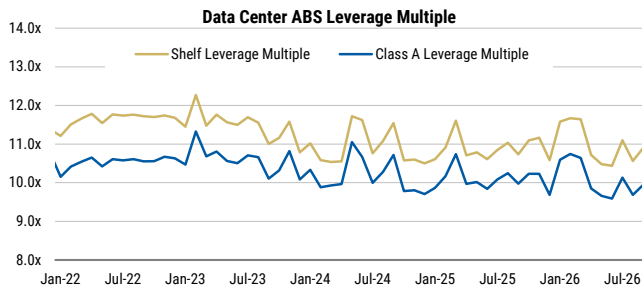
Source: Intex, Morgan Stanley Research

Exhibit 81: Data Center ABS Loan-to-Value Ratios



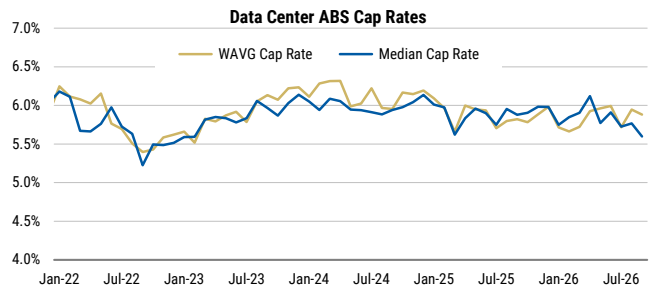
Source: Intex, Morgan Stanley Research

Exhibit 82: Data Center ABS Leverage Multiples



Source: Intex, Morgan Stanley Research

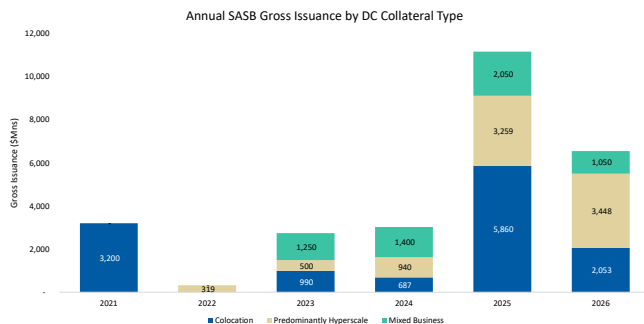
Exhibit 83: Data Center ABS Cap Rates



Source: Intex, Morgan Stanley Research

Data Center CMBS New Issue Credit Metrics & Pricing

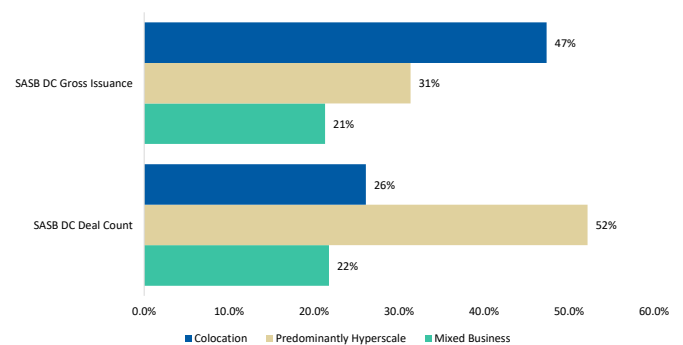
Exhibit 84: SASB data center YTD issuance has reached \$6.6bn across 6 transactions with another deal currently being marketed



Source: Trepp, Morgan Stanley Research

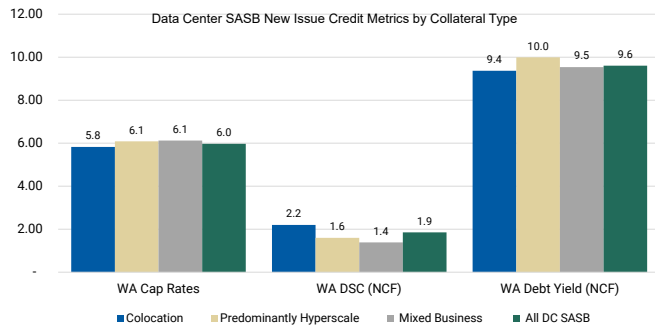
Exhibit 85: Predominantly hyperscaler-backed deals lead by count; colocation leads by issuance volume

Composition of SASB Data Center Issuance by Collateral Type



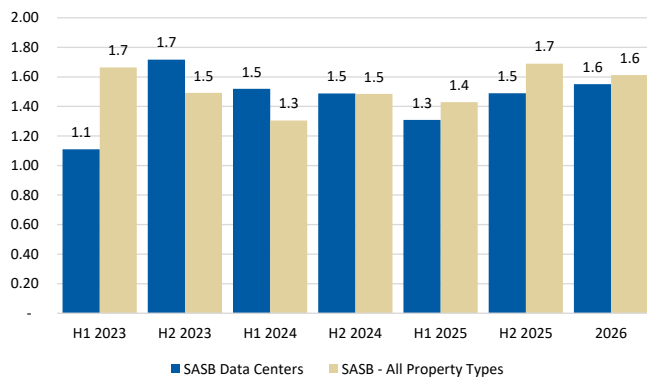
Source: Trepp, Morgan Stanley Research

Exhibit 86: Colocation DC SASBs have the highest weighted-average debt-service coverage



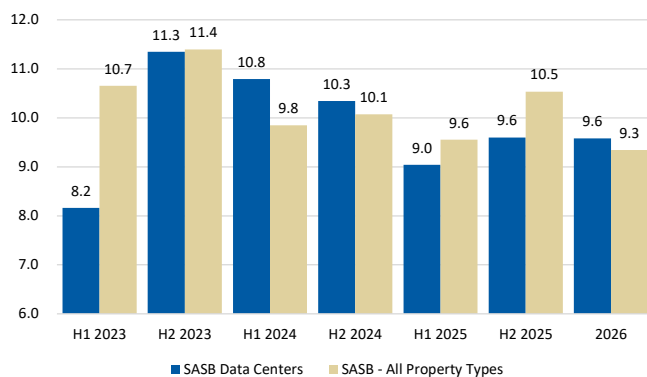
Source: Trepp, Morgan Stanley Research

Exhibit 88: The WA debt service coverage ratio for 2026 DC SASBs is ~1.6x, broadly consistent with that of all SASB new issues



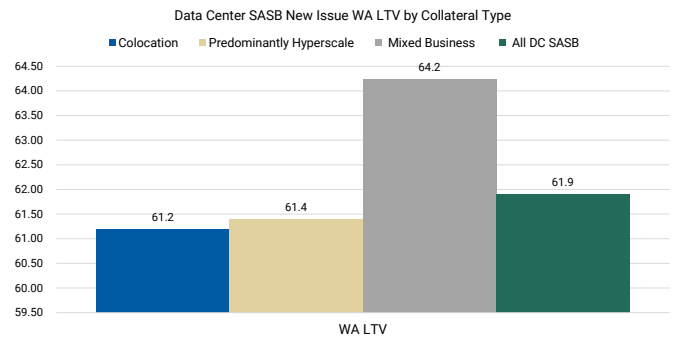
Source: Trepp, Morgan Stanley Research

Exhibit 90: Debt yields have trended lower across 2026 data center SASB transactions relative to H2 2025 levels; though recent deals report healthy debt yields in the 9%-15% range



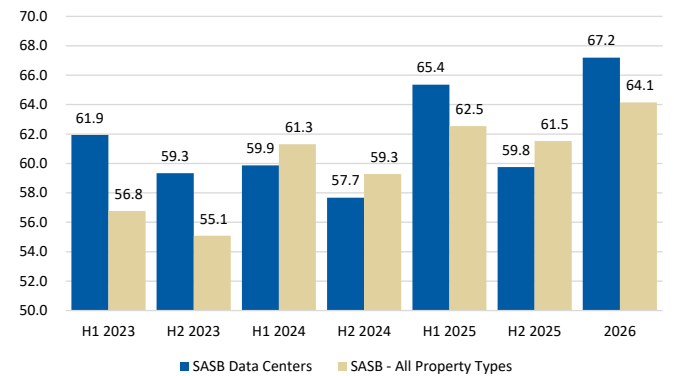
Source: Trepp, Morgan Stanley Research

Exhibit 87: Mixed business reports the highest leverage among DC collateral types



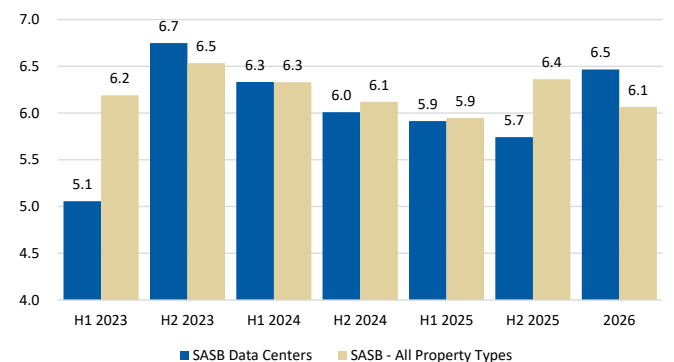
Source: Trepp, Morgan Stanley Research

Exhibit 89: Average LTV for 2026 data center new issue SASBs have hovered ~67.2%, above the broader SASB average

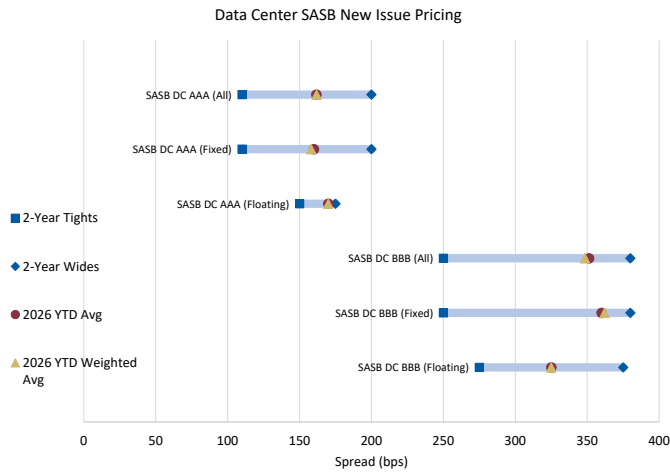


Source: Trepp, Morgan Stanley Research

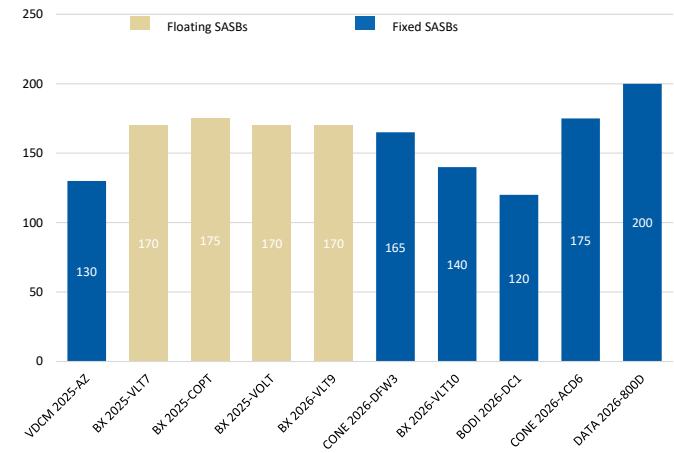
Exhibit 91: Cap rates on 2026 data center SASBs have widened to ~6.5% vs 2025 vintages



Source: Trepp, Morgan Stanley Research

Exhibit 92: New issue SASB DC Spreads - YTD average vs 2yr tight and wides

Source: Green Street Advisors; Morgan Stanley Research

Exhibit 93: DC SASB New Issue AAA Spreads

Source: Trepp, Morgan Stanley Research

Data Center ABS and CMBS Recent Pricings

Exhibit 94: New issue data center ABS

ABS New Issue Market													
Deal	Tranche	CUSIP	Pricing Date	ARD	WAL	Rating	Original Size	Current Size	Year	Spread	Coupon	Current YTM	LTV at Issuance
VDC 2026-1	VDC 2026-1 A2-I	92212KAL0	9/11/2026	3/15/2032	5.5	AAA	225	225	2026	140	6.13%	6.41%	26.10%
VDC 2026-1	VDC 2026-1 A2-II	92212KAM	9/11/2026	3/15/2032	5.5	AA-	110	110	2026	180	6.52%	6.81%	38.20%
VDC 2026-1	VDC 2026-1 A2-III	92212KAN6	9/11/2026	3/15/2032	5.5	A-	205	205	2026	200	6.72%	7.00%	60.60%
QTSII 2026-8	QTSII 2026-8 A2-I	74690FBU9	9/30/2026	10/5/2031	5.0	AAA	320.5	320.5	2026	150	6.44%	6.53%	42.60%
QTSII 2026-8	QTSII 2026-8 A2-II	74690FBV7	9/30/2026	10/5/2031	5.0	AA-	57.5	57.5	2026	200	6.93%	7.03%	49.90%
QTSII 2026-8	QTSII 2026-8 A2-III		9/30/2026	10/5/2031	5.0	A	162	162	2026	250	7.12%	7.53%	60.60%
QTSII 2026-8	QTSII 2026-8 B	74690FBX3	9/30/2026	10/5/2031	5.0	BBB-	130	130	2026	355	8.38%	8.58%	74.30%
QTSII 2026-9	QTSII 2026-9 A2-II	74690FBZ8	9/30/2026	10/5/2033	7.0	AA-	40	40	2026	215	7.15%	7.28%	49.90%
QTSII 2026-10	QTSII 2026-10 A2-III	74690FCC8	9/30/2026	10/5/2031	5.0	AAA	215	215	2026	SOFR + 175			42.60%
QTSII 2026-11	QTSII 2026-11 A2-III		9/30/2026	10/5/2030	4.0	AAA	75	75	2026				42.60%

Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research

Exhibit 95: New issue data center CMBS

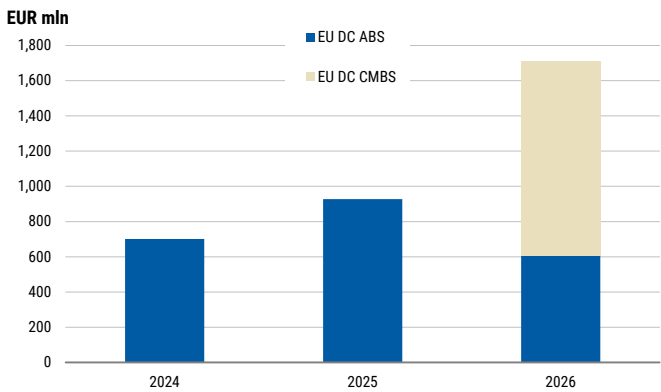
SASB New Issue Market											
Deal Name	Tranche	Cusip	Pricing Date	Class	Tranche Size (\$mn)	Spread	Coupon	Rate Type	Current Mid YTM	Min Original Rating	
DATA 2026-800D	DATA 2026-800D A Mtge	TC810401	9/8/2026	A	\$321.30	200	6.42%	Fixed	6.85	AAA	
DATA 2026-800D	DATA 2026-800D B Mtge	TC810402	9/8/2026	B	\$35.00	260	7.00%	Fixed	7.43	AA	
DATA 2026-800D	DATA 2026-800D VRR Mtge	TC810403	9/8/2026	VRR Int.	\$18.80			Fixed			

Source: Bloomberg, Trepp, Morgan Stanley Research

European DC ABS & CMBS

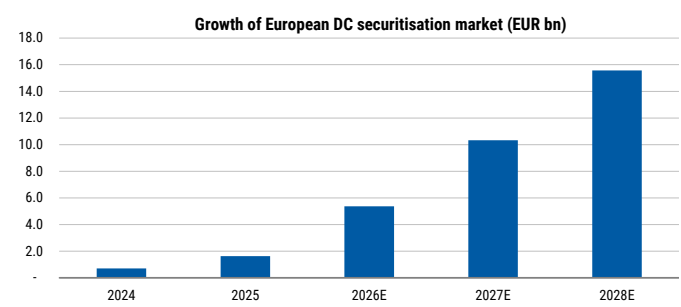
The European data centre securitization space is small for now, comprising a little over €3bln in current outstanding size from three ABS transactions and two CMBS deals. The absolute numbers are small but the momentum seems to be picking up - three new data centre operators have issued a transaction this year. Issuance in 2026 aggregates €1.7bln so far and has included the sector's first CMBS structures as well.

Exhibit 96: European DC securitized issuance is modest but growing



Source: Bloomberg, Morgan Stanley Research. Note: Our CMBS numbers include the Equinix xScale deal which has been issued as a secured bond.

Exhibit 97: The European DC ABS/CMBS market is small for now but we expect it to grow too as capacity expands



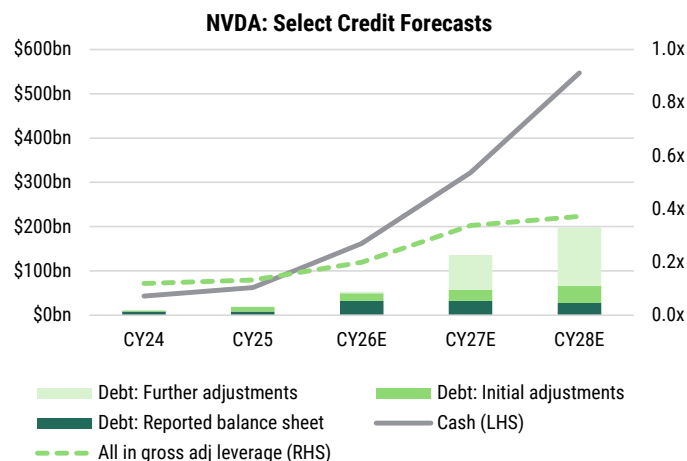
Source: Bloomberg, Morgan Stanley Research forecasts

Single Name Coverage: What's Topical Of Late in IG TMT

Our single-name technology credit work remains focused on AI implications, including financing structures, ecosystem support, and supply-demand dynamics. Despite lighter September issuance, off-cycle earnings and headline flow continue to drive elevated investor inbound. NVDA and ORCL, both within our current coverage, remain central to these discussions, from a funding and lease documentation perspective:

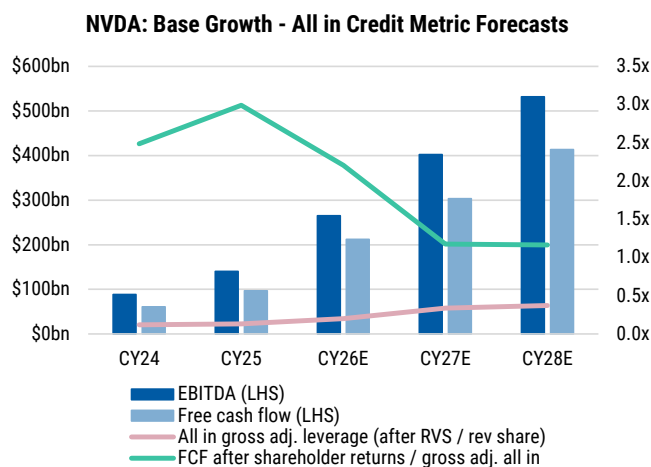
- **NVDA – funding update:** We [initiated NVDA credit coverage](#) with a neutral view. Exceptional growth has turned the balance sheet into a strategic financing tool, but the potential tail risks of what we call "balance sheet-as-a-service" remain early-stage, opaque, and significant. The [FY2Q27 print](#) reinforced an expanding support toolkit that now includes shell lease guarantees, revenue-share/contract price-floor support, and potential RVS structures tied to the announced \$500bn partnership. A key debate remains how markets absorb ~\$1tn guided GPU/XPU-related financing, including the structures, ratings treatment, and investor bases.
- **NVDA – lease documentation development:** With its FY2Q27 earnings, NVDA's disclosure of plans to reassign ~\$20bn of future shell leases intensified investor scrutiny of powered-shell documentation.
- **ORCL – funding update:** ORCL completed its \$20bn equity ATM program in FY1Q27. We [forecast sizable bond issuance](#) in early CY27 and CY28. Customer prepaes have emerged as a meaningful funding source, albeit with limited disclosure and associated rollover sensitivity. Future equity needs likely depend on prepay success and cost inflation. Lease disclosures imply [a ~1GW shell lease signing](#) during the quarter. We look for greater clarity on long-term funding, capex, and build-out plans at the October Analyst Day.
- **ORCL – lease documentation development:** Bloomberg reported on September 24 that ORCL issued a force majeure notice to the developer of Project Jupiter, though Oracle stated on X the project remains on schedule, and such notices are commonplace tools used to preserve contractual rights. The surprising development prompted [this response across a few Research teams](#).
- **ORCL – construction financing link:** Given heightened attention on Stargate sites, we highlight ~\$65bn of short-dated bank and private-credit construction loans funding shells to be leased by ORCL, based on company disclosures and Bloomberg news articles including one dated 4/24/26. The ~\$65bn figure sits outside the public IG and HY construction debt discussed elsewhere in this note by our Strategy colleagues and, accordingly, excludes RDMICH.

Exhibit 98: NVDA: In our credit initiation in late August, we presented a forecast for CY28-end all-in credit exposure scaling to ~\$200bn



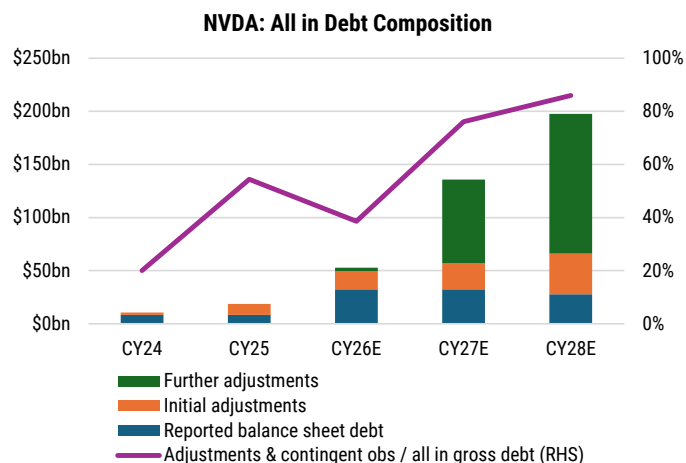
Source: Company filings, Morgan Stanley Equity Research (Moore model), Moody's, S&P, Bloomberg, Morgan Stanley Credit Research. Read more: [TMT Credit Research: NVDA Credit: Balance Sheet-as-a-Service](#) (24 Aug 2026) and [NVIDIA Corp.: Credit & Compute | Assessing NVDA's 'Balance Sheet-as-a-Service' From the Cross-Asset Lens](#) (9 Sep 2026).

Exhibit 100: NVDA: However, credit metrics look strong even as contingent exposure scales and even if headline growth moderates



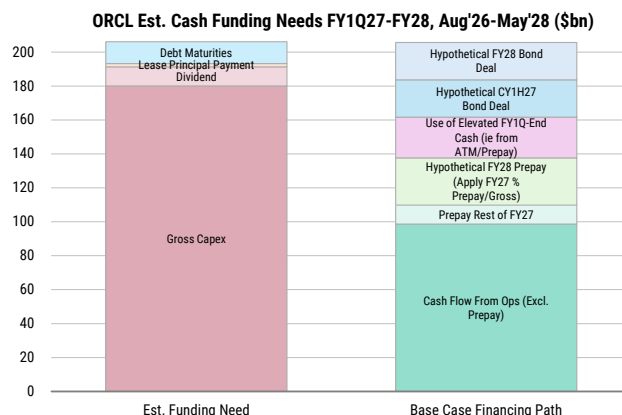
Source: Company filings, Morgan Stanley Equity Research (Moore model), Moody's, S&P, Bloomberg, Morgan Stanley Credit Research

Exhibit 99: NVDA: That estimate features ~\$170bn away from funded debt, largely driven by our forecasting tied to the \$500bn partnership and the revenue-share program



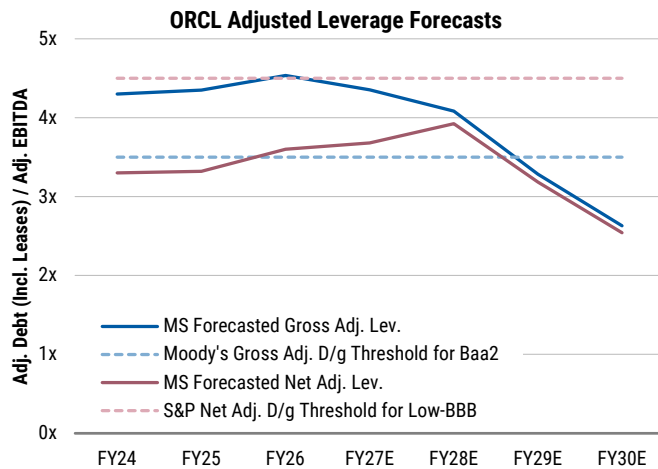
Source: Company filings, Morgan Stanley Equity Research (Moore model), Moody's, S&P, Bloomberg, Morgan Stanley Credit Research

Exhibit 101: ORCL: After the FY1Q27 print, we think parent funding needs may require two large bond deals by FY28-end even if ORCL receives ~\$40bn more customer prepayments



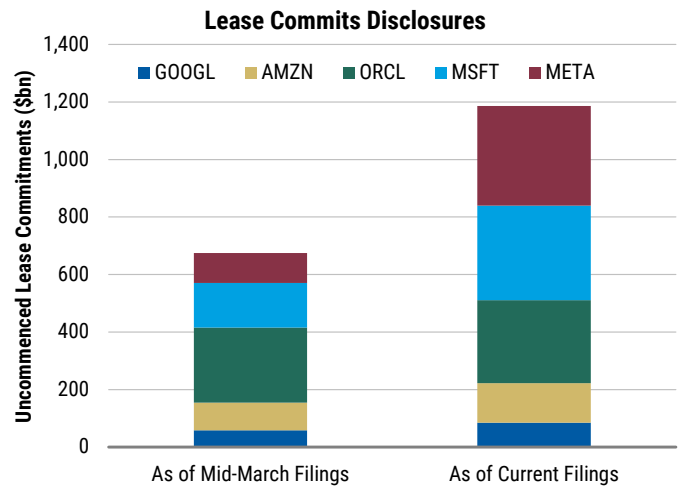
Source: Company filings, Morgan Stanley Credit Research, Morgan Stanley Equity Research. Read more: [TMT Credit Research: ORCL Credit: New Bookings, Old Bills - FY1Q27 Quick Takes](#) (11 Sep 2026).

Exhibit 102: ORCL: After the FY1Q27 print, ratings risk remains, with our forecasted gross adj. lev. >3.5x through FY28-end and potential for further pressure in FY29-30 given consensus capex runs above the levels reflected here



Source: Company filings, Moody's, S&P (note lev forecasted higher at 4.4x), Morgan Stanley Credit Research, Morgan Stanley Equity Research

Exhibit 103: ORCL: Lease commitments now exceed \$1.1tn across major IG hyperscalers, with ORCL no longer the largest contributor despite a reacceleration in FY1Q27



Source: Company filings, Morgan Stanley Research. Read more: [TMT Credit Research: ORCL Credit: Off Balance Sheet, On the Rise – FY1Q27 10-Q Highlights \(11 Sep 2026\)](#).

ORCL Trade Valuation Methodology and Risks - Lindsay Tyler

TRADE	ENTRY LEVEL	ENTRY DATE	CURRENT LEVEL	RATIONALE	RISKS
Buy ORCLCP 5Y CDS	+57bp	9/28/25	+244bp	• Credit profile deterioration, driven by cash funding needs and increasing lease liabilities related to new data centers • Lender hedging may be a key driver of CDS too • For relative value, we've considered other large structure situations where there have been ratings risk	• Constructive updates on RPO/growth and financing clarity • TL As may not ultimately serve a significant funding role • CDS technicals: Basis selling, construction loan hedge unwinds, fast money outright shorts profit taking
Sell ORCL 5.2 9/26/35	+167bp	1/23/26	+212bp	• Credit profile deterioration, driven by cash funding needs and increasing lease liabilities related to new data centers • For relative value, we've considered large media/cable structure comps	• Additional visibility on long-term monetization and delevering beyond FY28-end • Issuance cadence may be different than our assumptions
Sell ORCL 5.95 9/26/55	+208bp		+264bp		

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(as of September 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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